RESUME OS

Master Intelligence System — vFinal Unified

Three-Layer Canonical Specification

Layer 1: Philosophy + Research Foundation | Layer 2: System Architecture | Layer 3: Execution Specification

What this document is: The single canonical source of truth for Resume OS. It is not a strategy document. It is not a formatting guide. It is a fully executable OS-level specification for a career signal engineering system. Every rule has an input contract, output contract, research source, and failure mode.

How to read it: Layer 1 (Philosophy + Research) establishes why every rule exists. Layer 2 (System Architecture) defines what the system does. Layer 3 (Execution Spec) defines how it is built and run. All three layers are required — none is optional.

Epistemic discipline: Every claim is tagged FACT (named study, verified), INFERENCE (logically derived), or ASSUMPTION (hypothesis pending outcome validation). Tags are not decorative — they define which rules can be treated as engineering constraints vs design hypotheses.

Completeness state (March 2026): Conceptual: 95% | Structural: 90% | Executable: 90% | Validated by outcome data: ~15% (pilot underway, minimum 50 samples per cluster-level needed for learning layer activation)

What is still open: University DNA configuration spec | Persona Match definition (Pick Hire) | Cover letter scope decision | LLM parser optimisation rules | Emerging role variants in Decision Engine | Simulation/Output Validation Layer (Layer 3, Section 3.7 — new in this version)

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LAYER 1

Philosophy + Research Foundation

Why every rule exists. The evidence base. Epistemic discipline.

1.1 THE CENTRAL PROBLEM

Talent is not missing. It is invisible to the systems that decide who gets hired. The resume is the signal layer between a candidate's capability and a recruiter's decision. Most resumes fail not because the candidate lacks ability — but because the signal is misrepresented, misformatted, or mistargeted. Resume OS corrects all three.

Every existing resume tool in market treats the resume as a document to be formatted. Resume OS treats it as structured career data to be engineered — then rendered into a format-specific output. The resume is not the product. The structured signal model is.

`SOURCE: v1.0 Sec 0 + v2.0 Sec 0 + v3.0 Governing Principle + Research Doc (all 7 domains)`

1.2 THE EVIDENCE BASE — RESEARCH PILLARS

> *Tagging convention used throughout this document: FACT = named study, verified methodology, confirmed finding. INFERENCE = logically derived from one or more FACT sources. ASSUMPTION = hypothesis pending validation by outcome data. Tags are not decorative — they determine whether a rule is an engineering constraint (FACT/INFERENCE) or a design hypothesis (ASSUMPTION).*

Research Pillar	Key Studies + Sample Sizes	Epistemic Tag	System Application
Recruiter scan behaviour	TheLadders 2018 (n=30, eye-tracking, 7.4s triage). Tegze 2023 (n=114, 17–46s). Interviewing.io 2024 (n=76 recruiters, ~2,200 evals, 31s median). ResumeGo 2024 (n=418, self-report).	FACT	Three-window temporal model. Triage Zone definition. Dual-phase optimisation.
Scanning patterns	NNG, Nielsen 2006 (F-pattern, n=232). NNG, Pernice 2019 (Layer-Cake, Spotted, Bypassing). Valor Software (I-pattern). NNG (Commitment, Marking).	FACT	Pattern Intelligence Layer (Layer 0). Structure Engine. Layer-Cake as target state.
Recruiter decision accuracy	Interviewing.io/Learning Collider 2024 (~1,000 resumes cross-validated vs mock interviews). Recruiters: 55% accuracy. SD across candidates: 0.34 vs same candidate: 0.32.	FACT	Trust Signal placement priority. Pedigree heuristic engineering. Cognitive Signal as System 2 activation mechanism.
Cognitive load theory	Cowan 2001 (4±1 chunks, 5,600+ citations). Chandler & Sweller 1991 (split-attention effect). Thalmann et al. 2019 (chunking + LTM).	FACT	Bullet count limits (3–5 per role). Section density rules. Cognitive Load Score calculation.
Serial position effects	Murdock 1962 (n=103, U-shaped recall curve). Glanzer & Cunitz 1966 (n=240, primacy via LTM, recency via WM).	FACT	Bullet ordering rules. Primacy rule: bullet #1 = strongest Hard Signal. Recency rule: bullet #last = second strength.
Pre-attentive processing	Treisman Feature Integration Theory 1980s. Colin Ware Information Visualization 2004. Willis & Todorov 2006 (Princeton, 5 experiments, 100ms judgment formation).	FACT	Pre-Attentive Design Layer. Whitespace rules. Typographic hierarchy. Spotted Pattern trigger design.
ATS parsing systems	EDLIGO 2025 (1,000 resumes, Workday/Taleo/Greenhouse — 43% non-qual rejections). Jobscan 2025 (99.7% keyword filter usage, n=384). Enhancv 2025 (n=25 recruiters, 8% auto-reject, 92% rank/sort). Harvard/Accenture 2021 (88% employer signal loss).	FACT	ATS Intelligence Layer. Parsing Gate. Platform-specific rules. ATS Compatibility Score.
Document design science	Chaparro et al. 2004 (Wichita State, n=20, F(1,17)=8.34, p=.01). Rello & Pielot 2016 (n=104, line spacing). Alex Poole 2008 (serif vs sans-serif, 50+ studies). TalentWorks (475–600 word optimum, n=6,305).	FACT	Whitespace ratio rules. Interline spacing 1.15–1.3. Font compliance. Word count targets.

India campus hiring	AICTE 2024 (14.90 lakh B.Tech seats). CII+Wheebox+AICTE 2025 (71.5% employability). India Skills Report 2024 (Telangana: 85.45% employability, 68.71% placement). Deloitte India 2024 (n=190+ orgs, 500+ campuses, 93% weight on technical interview).	FACT	Fresher Proxy Signal Architecture. India ATS stack. PickCV assessment integration. AP/Telangana context.
AI + LLM bias in hiring	UW 2024 (85.1% racial bias, n=9 occupations). Meng et al. 2024 AAAI (white-associated names 85.1%). PMC/NIH 2025 (36,000+ resumes, 5 LLMs). Wilson et al. 2025 (AAAI, n=528).	FACT	Bias Mitigation Engine (Sub-Engine 5.9). Bias Audit Flag per variant.
Metrics + conversion	TalentWorks (40% interview lift from quantified achievements, n=6,305). Cultivated Culture (36% zero metrics, 26% have 5+, n=125,000+). Wonsulting eye-tracking (80% bullets should have numbers). ResumeGo (LinkedIn URL = 71% lift).	FACT	Hard Signal minimum enforcement. Signal Density Score. Bullet formula requirements.
Employment gaps + bias	Harvard Kennedy School RCT (n=9,022 employers + 2,650 lab, 15% callback lift with years-of-experience format). Pew Charitable Trusts (6+ month gap = 3.5× more resumes needed).	FACT	Gap-neutral date formatting option. Bias Mitigation Engine.
Environment adaptation	Derous & Ryan 2019 HR Management Review (hiring bias by org type).	INFERENCE	Environment Adaptation Engine. Startup/SME/MNC signal framing.
Pattern → outcome mapping	No empirical validation. Theoretically derived from role-specific recruiter behaviour research.	ASSUMPTION	Variant recommendation by cluster. Requires 50+ samples per cluster-level to validate.

SOURCE: Research Doc (all 7 domains) + v2.0 Sec 0.2 (Research Foundation table) + v3.0 Sec 1 (evidence base)

1.3 THE THREE-WINDOW TEMPORAL MODEL

> *CORRECTION from v1/v2: The 6-second and 7.4-second models have been superseded. The Interviewing.io 2024 study (n=76, ~2,200 evaluations) established a 31-second median. Three distinct windows exist — they are nested, not sequential. Failure at 100ms prevents reaching 7 seconds.*

Window	Duration	Cognitive Process	Primary Optimisation Target	Failure Mode
Pre-attentive	50–500ms	Parallel feature processing: density, hierarchy, alignment, rhythm. Competence/trustworthiness judgment formed (Willis & Todorov 2006). Cannot be reversed by later content.	Visual quality: whitespace ratio ≥20%, 3-level typographic hierarchy, left-margin alignment consistency, Calibri or Tier-1 font.	Multi-column, icon-heavy, dense text → negative competence judgment before a word is read
Triage scan	7–15 seconds	Heuristic pattern-matching. Six critical fixation anchors (TheLadders 2018): name, current title/company, previous title/company, current dates, previous dates, education.	Six data points visible in top 40% (Triage Zone). Trust Signals (company names) bold. Strongest Hard Signal in bullet #1 of current role.	Any of the 6 data points absent from Triage Zone → Fit/No Fit decision made on incomplete signal

Evaluation read	25–45 seconds	Analytical (System 2) IF cognitive load is low. Heuristic (System 1) if overloaded. Accepted: avg 25s. Rejected: avg 44s — longer rejection when triage signals are ambiguous.	Achievement depth. Cognitive load budget per section (≤5 bullets). Serial position ordering. Self-contained bullets. No split-attention.	Cognitive overload forces recruiter back to heuristic mode — 55% accuracy baseline (Interviewing.io 2024)
ATS parsing	Milliseconds (automated)	Rule-based machine parsing. Ranks, does not primarily auto-reject (8% configured for auto-reject, Enhancv 2025).	Parsing Gate compliance. Keyword density. Platform-specific formatting.	Parsing Gate failure → all keyword and relevance scoring invalidated regardless of content quality

SOURCE: TheLadders 2018 + Interviewing.io 2024 + Willis & Todorov 2006 + Enhancv 2025 + v3.0 Sec 1.1

1.4 WHY RESUME OS CANNOT BE JUST A DOCUMENT EDITOR

The core failure of every competitor: they treat the resume as a document to be formatted rather than structured career data to be engineered. No competitor builds a talent graph. None converts career narratives into queryable, comparable, computable signals. All operate on keyword overlap — none model semantic signal weight or how recruiter cognition processes information.

- Jobscan: ATS keyword optimisation only. No cognitive science layer. No signal architecture. No India-specific logic.
- Resume Worded: Transparent scoring but document-centric. No structured data model.
- Rezi/Teal/Enhancv: GPT-based content generation with templates. No signal classification.
- GY Campus (India): Assessment-first, no resume intelligence. The gap between GY Campus and Jobscan is exactly where Resume OS operates.
- Naukri Pro AI Resume Maker (Nov 2025): Templates not fully ATS-friendly. AI documented errors. No cognitive science layer.

74% of HR professionals can spot AI-generated resumes; 57% are less likely to hire those applicants (Resumly 2025). Current tools optimise for machines at the expense of human authenticity. Resume OS engineers for both simultaneously.

SOURCE: Research Doc Domain 7 (competitor analysis) + v3.0 Sec 6.3 (business moat)

LAYER 2

System Architecture

What the system does. Every layer. Every component. Every rule.

2.1 FULL PROCESSING PIPELINE

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INPUT: Master Resume (any format) + JD text OR Job URL

PRE-GATE: Constraint Engine
  HARD FAIL → Multi-column + Workday/Taleo | Tables in body | Keyword coverage <70%
  HARD FAIL → No metrics in first 2 bullets | >5 bullets per role
  SOFT FAIL → Passive language >30% | No leadership verbs at L3+ | Orphaned keywords

LAYER 0:  Pattern Intelligence    → Pattern (V1-V10) + scanning strategy
LAYER 1:  Role Intelligence       → Role DNA: Function + Level + Environment + KPIs
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LAYER 2:  ATS Intelligence        → ATS platform + formatting rules + file format
LAYER 2.5: Signal Architecture    → Hard/Structural/Cognitive/Trust classification index
LAYER 3:  Recruiter Decision Model → Triage risk + pattern activation + risk flags
LAYER 4:  Gap Analysis            → Gap Scorecard (5 dimensions)
LAYER 5:  Optimisation Engine     → 9 sub-engines in sequence
LAYER 6:  Output Engine           → 7-10 variants + scores + Strategy Log
LAYER 7:  Simulation/Validation   → [NEW] Recruiter decision simulation + output validation

OUTPUT PACKAGE: Variants + Score Table + Default Recommendation + Strategy Log
               + Gap Notification + ATS Advisory + Bias Audit Flag + Sim Report
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SOURCE: v1.0 Sec 1 + v2.0 Sec 1 + v3.0 Sec 4.1 + v3.3 Sec 1 + Layer 7 new in vFinal Unified

2.2 ROLE DNA — THE FOUR AXES

Every Resume OS run begins with Role DNA. Without it, optimisation defaults to generic keyword stuffing — the least effective intervention. Role DNA is the input to every subsequent layer.

Axis	Definition	Values	Effect on System	Failure if Missing
FUNCTION	What the role does — functional domain	GTM (Sales, Marketing, CS) | Tech (Engineering, Product, Data) | Business (Ops, Strategy, HR) | Support (Finance, Legal, Admin)	Selects role cluster (C1–C10) and its bullet formula, ATS priority, and keyword set	Generic keyword list, wrong bullet formula, wrong pattern selected
LEVEL	What the role expects the person to prove	L1–L2 Execution (0–2yr) | L3 Ownership (3–6yr) | L4 Strategic (7–12yr) | L5+ Business Impact (12+yr)	Determines resume bias, summary length, bullet framing, section ordering, leadership language expectation	Level mismatch — most common failure mode. 5-year candidate gets L2 language. Fails heuristic pattern-match before recruiter reads a bullet.
ENVIRONMENT	Company type and cultural hiring bias	Startup (0–200 emp: speed + ownership) | SME (200–2,000: versatility) | MNC (2,000+: process + specialisation)	Sets tone of every bullet. Amplify/suppress language pairs. Framing of all achievements.	Wrong cultural fit signal — startup candidate using MNC governance language signals misalignment
KPIs	How success is measured — extracted from JD	Extracted: quota, uptime, TTF, CAC, NPS, ARR, etc. Inferred from role × level if not explicit.	Drives metric framing in bullets. Determines which quantified outcomes are most relevant to surface.	Generic metric framing not aligned to what the role values

SOURCE: v1.0 Sec 2 + v2.0 Sec 3 + v3.0 Sec 4.1 + Research Doc Domain 1

2.3 SIGNAL ARCHITECTURE — THE FOUR TYPES

Type	Definition	Examples	Weight	Placement Rule	ATS Impact
HARD	Quantified, verifiable outcomes. The only signal type that survives both ATS filtering and human scepticism simultaneously.	Revenue: ₹2.8Cr ARR | Quota: 134% | Users: 800K | Model: 94.2% vs 87.1% baseline | Latency: 340ms→42ms	40%	First 1–2 bullets of every role. Summary line 2. Triage Zone mandatory.	Contributes to Experience Relevance (20% of ATS score) when metric language aligns with JD KPI vocabulary
COGNITIVE	Evidence of thinking quality via problem→diagnosis→action→outcome structure. NOT claimed through adjectives —	Problem identified → root cause diagnosed → specific intervention → quantified result. Demonstrates	25%	Current + most recent prior role bullets 1–3. NOT in summary. NOT claimed — demonstrated.	Indirect: cognitive signal bullets with JD keywords contribute to Experience Relevance. No direct ATS pathway.

	demonstrated through bullet architecture.	analytical thinking without claiming it.			
STRUCTURAL	Signal conveyed by resume architecture itself — section ordering, bullet density, visual hierarchy, whitespace. Implicit — emerges from discipline, not content.	Consistent section ordering | Logical bullet hierarchy | Appropriate recency weighting | Clean typography | Absence of noise	20%	Entire document. Most critical in Triage Zone. Pre-attentive processing evaluates structural signals before content.	Determines Parsing Gate pass/fail. Parsing Gate failure invalidates ALL other ATS score components.
TRUST	Third-party verification and institutional credibility. Bypasses scepticism applied to self-reported claims. Activates pedigree heuristic.	FAANG/top-firm name (35% interview rate lift, Interviewing.io 2024) | University | Certification + issuer ID | GitHub active repo | Portfolio with live deployment | Assessment score + verifier	15%	Primary Optical Area (top-left, above fold). Company names bold in experience block. Verification markers adjacent to claims.	Does NOT contribute to ATS Compatibility Score. ATS systems do not weight company prestige. Trust Signals are exclusively for human cognitive processing.

> *CONFLICT NOTE: Signal Architecture weights (40/25/20/15) and ATS Compatibility Score components (Keyword 35% + Title 25% + Experience 20% + Skills 10% + Parsing Gate) are parallel measurement systems. They are NOT directly mapped. A high Trust Signal density does not improve ATS score. A high ATS score does not require Cognitive Signals. Both scores are reported independently per variant.*

SOURCE: `v3.0 Sec 3 + v3.3 Sec 4 + Patch C (vFinal PATCH doc)`

2.4 SIGNAL PRIORITY STACK BY CLUSTER

Cluster	P1 (Dominant)	P2	P3	P4 (Contextualise/Suppress)
C1 SWE/Full-Stack	STRUCTURAL (stack depth, exact tool names)	HARD (system scale: users, throughput, latency)	TRUST (GitHub URL, known companies)	COGNITIVE (architecture rationale — save for commitment phase)
C2 Data/BA	HARD (business outcome per analysis)	COGNITIVE (insight→decision chain)	STRUCTURAL (tool + scope clarity)	TRUST (institution, certification)
C3 DS/ML	HARD (model accuracy vs baseline — mandatory)	TRUST (publications, competition rank)	COGNITIVE (experiment design, hypothesis)	STRUCTURAL (framework + deployment)
C4 Product	COGNITIVE (problem→decision→outcome)	HARD (product metrics: DAU, ARR, NPS)	TRUST (company name, product scale)	STRUCTURAL (cross-functional scope)
C5 DevOps/SRE	STRUCTURAL (exact infra stack)	HARD (uptime %, cost reduction, automation)	TRUST (cloud certifications with ID)	COGNITIVE (architectural rationale)
C6 Sales/BizDev	HARD (quota%, ARR/ACV, deal size — every bullet)	TRUST (company name, named accounts)	STRUCTURAL (CRM proficiency)	COGNITIVE (sales methodology)
C7 Marketing/Growth	HARD (campaign ROI, CAC, ROAS, conversion)	STRUCTURAL (channel + budget scope)	COGNITIVE (experiment design, lift)	TRUST (agency, brand, industry)
C8 HR/TA	HARD (hiring volume, TTF, offer acceptance rate)	STRUCTURAL (HRTech stack)	TRUST (company, programme scale)	COGNITIVE (system design thinking)

C9 Ops/Consulting	COGNITIVE (SAR: situation→action→result)	HARD (efficiency gain, cost reduction, scale)	STRUCTURAL (process + methodology)	TRUST (client names, certifications)
C10 UX/UI Design	TRUST (portfolio URL — hero above fold)	COGNITIVE (design process: problem→research→decision→outcome)	HARD (user outcome metrics)	STRUCTURAL (tool proficiency)
Fresher (all clusters)	TRUST (verified score, institution, GitHub)	STRUCTURAL (project scope + tech stack)	HARD (quantified project proxies)	COGNITIVE (learning velocity)

SOURCE: v3.0 Sec 3.2 + v3.3 Sec 4 + Patch A Engine 0

2.5 ATS PLATFORM INTELLIGENCE

Platform	URL Signature	Parsing Behaviour	File Format	Critical Rules
Greenhouse	boards.greenhouse.io	Keyword + structured. PDF parses well. Keyword density in first half critical.	PDF preferred	Keyword density front-loaded. Portfolio/GitHub links carry weight.
Lever	jobs.lever.co	Keyword + slight semantic. More forgiving of structure variation. Favours GitHub + portfolio.	PDF preferred	Portfolio and GitHub carry weight. Semantic variants supported.
Workday (39% of Fortune 500)	myworkdayjobs.com	Heavy filtering. Rigid section parsing. Strict standard headers required.	DOCX (4% failure vs 18% for PDF)	Standard headers mandatory: Education, Experience, Skills. DOCX significantly more reliable.
Taleo	taleo.net	Rigid keyword. Most sensitive to formatting.	DOCX / plain text	No tables. No columns. No special characters in section headers. Most restrictive platform.
Ashby	ashbyhq.com	Slight semantic. Most modern. Recognises skill synonyms.	PDF preferred	Portfolio and GitHub carry weight. Synonym recognition makes exact-match less critical.
SAP SuccessFactors	successfactors.com	Enterprise-grade filtering. Exact tool names critical.	DOCX preferred	'Power BI' not 'PowerBI'. Section order matters.
SmartRecruiters	smartrecruiters.com	Keyword + skills. Strong skills section weighting. Certifications parse well.	DOCX preferred (enterprise)	Skills section weighting higher than most platforms.
Naukri RMS (India — dominant)	No URL detection; must be specified	Primarily keyword-based. 78.3M resume database. Basic regex/RChilli parsing.	DOCX or PDF both	Exact-match optimisation mandatory. No semantic matching. Most common campus hiring platform in India.
Darwinbox (India)	No URL detection	Structured. Standard header compliance required. More sophisticated than Naukri.	DOCX preferred	Mid-to-large enterprise focus. Standard section headers critical.
Keka / Freshteam / GreytHR (India)	No URL detection	Basic ATS function. Keyword exact-match.	DOCX or PDF	Aggressive exact-match keyword coverage needed. Minimal semantic capability.
Unknown ATS	None detected	Apply generic ATS-safe rules.	PDF (default)	Single column, standard headers, no tables in body. Report 'Unknown ATS — generic rules applied'.

SOURCE: v1.0 Sec 3 + v2.0 Sec 4 + v3.0 ATS Intelligence Layer + Research Doc Domain 2 (India ATS)

2.6 THE 10 OUTPUT VARIANTS

Variant	Target Context	ATS	Section Order	Tone + Signal Emphasis
V1 Signal Stack	Tech: SWE, DevOps, ML at product companies	Greenhouse, Lever	Skills → Projects → Experience → Education	Technical, direct, evidence-led. Stack depth first.
V2 Outcome Ledger	PM, Sales, Marketing — outcome-driven roles	Greenhouse, Workday	Summary → Metrics Block → Experience → Skills → Education	Results-first. Numbers in triage zone. Business language.
V3 Authority Frame	Enterprise/MNC: DA, BA, Ops, Consulting, HR	Workday, SAP SuccessFactors	Summary → Tools & Certifications → Experience → Education	Formal, process-led, governance-aware. MNC framing.
V4 Leadership Thesis	Senior/Lead roles: L4–L5. Manager→VP.	Any ATS	Exec Summary → Team/Scale Markers → Experience → Core Competencies → Education	Executive. Strategic. Individual contribution minimised.
V5 Proof Sheet	Research, ML, academic-industry hybrid	Lever, Ashby	Research Interests → Publications → Projects → Skills → Education (top)	Academic precision. Metric vs baseline mandatory.
V6 Problem-Solver	Ops, Consulting, Supply Chain, cross-functional	Workday, SmartRecruiters	Summary → Competency Matrix → Experience (SAR) → Certifications → Education	SAR format. Problem-solution language. Structured.
V7 Portfolio Lead	UX/UI Design, Product Design, creative roles	Lever, Ashby, Greenhouse	Portfolio URL (hero) → Summary → Case Studies → Tools → Education	Process-led, user-centric. Portfolio IS the resume.
V8 Versatility Map	SME multi-role, generalist positions	Greenhouse, Workday	Summary → Cross-functional Skills → Experience (breadth) → Education	Pragmatic, adaptable. Cross-functional language.
V9 Domain Expert	Deep specialist: Senior+ single domain	Greenhouse, Lever, Workday	Domain Summary → Industry Credentials → Experience (depth) → Publications/Certs → Education	Authority. Domain vocabulary throughout.
V10 Transition Narrative	Career changers, re-entrants, function pivots	Greenhouse, Lever	Transition Summary → Transferable Skills → Relevant Experience → New-Domain Projects → Education	Forward-looking. Explicit about pivot.

SOURCE: v1.0 Sec 7.2 + v2.0 Sec 8.2 + v3.0 Sec 4

2.7 SCORING SYSTEM — ALL 5 DIMENSIONS

Score	Weight in Composite	Calculation Method	Threshold System
ATS Compatibility	60% in default recommendation composite	Keyword Match 35% + Title Match 25% + Experience Relevance 20% + Skills Coverage 10% + Parsing Gate (pass/fail override). Platform-specific. Reported per variant with ATS name.	85–100%: Strong | 70–84%: Good | 55–69%: Fair | <55%: BLOCK — do not submit
Recruiter Scan	40% in default recommendation composite	Four triage checks evaluated in top 40% of document: Clarity + Relevance + Proof + Risk. 25-point scale each = 100 total.	Same threshold scale as ATS. Both scores reported separately.
Signal Density	Reported separately —	Hard sub-score 40%: bullets with metric ÷ total (position-weighted: bullets 1–2 count 2×, scope	No threshold block. Reported as diagnostic. Low Signal Density

	not in composite	proxies count 0.5×). Cognitive 25%: bullets with 3-part structure ÷ total in current + prior role. Structural 20%: binary checks (section order, headers, whitespace, hierarchy, alignment) ÷ 20. Trust 15%: presence of company bold + LinkedIn + GitHub/Portfolio + certification ID (capped 15).	with high ATS = document will rank but may not convert.
Cognitive Load	Reported separately — not in composite	Inverse measure. 6 parameters: whitespace ratio (20% threshold), bullet count compliance per recency band, words-per-bullet compliance (15–40), sections-per-page-1 (max 6), serial position adherence, bold element density (max 3 per bullet). Sum of passed checks ÷ 6.	Low Cognitive Load Score = high effort resume = likely to force heuristic mode. Reported as diagnostic.
Role Match	Not in composite. Informs Gap Scorecard.	Keyword Coverage% + Functional Relevance overlap. Keyword Coverage% = (JD required keywords present ÷ total JD required) × 100. Functional Relevance = proportion of JD action verbs present in experience bullets.	<70% keyword coverage → Keyword Engine full activation. <75% role match → Role-Shift Engine activation.

Default Recommendation Algorithm (deterministic):

```
1. Filter variants with ATS score < 55% → exclude from recommendation pool
2. If all variants < 55% → BLOCK: re-optimisation required before delivery
3. Composite Score = (ATS Score × 0.60) + (Recruiter Scan Score × 0.40)
4. Variant with highest Composite Score = Default Recommendation
5. Tie-break: prefer variant with higher ATS Score
6. Log rationale in Strategy Log
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SOURCE: v1.0 Sec 3.2 + v2.0 Sec 8.3 + v3.3 Sec 5 + Patch C scoring integration

3.0 COMPLETE SYSTEM PIPELINE — UPDATED

This diagram supersedes the pipeline in Section 2.1. It reflects all additions: Input Normalization, Trade-Off Engine, Authenticity Engine, Failure Diagnosis, Variant Distance Control, Fallback Modes, and the Simulation Layer.

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╔═══════════════════════════════════════════════════╗
║  INPUT GATE                                       ║
║  Master Resume + JD / URL + Optional preferences  ║
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║  FALLBACK MODE CHECK (Section 3.13)               ║
║  No JD → BLIND_MODE | No URL → GENERIC_ATS_MODE   ║
║  Low content → CONTENT_VOLUME_CORRECTION          ║
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║  INPUT NORMALIZATION ENGINE (Section 3.8) [NEW]   ║
║  Resume parse cleanup | JD structuring            ║
║  Metric inference tagging | Noise removal         ║
║  Consistency check → Normalized Input Package     ║
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║  CONSTRAINT ENGINE (Section 2.1)                  ║
║  Hard fail | Soft fail pre-checks                 ║
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L0: Pattern Intelligence  →  L1: Role Intelligence
                ↓
L2: ATS Intelligence      →  L2.5: Signal Architecture
                ↓
L3: Recruiter Decision Model → L4: Gap Analysis
                ↓
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║  L5: OPTIMISATION ENGINE (9 Sub-Engines)          ║
║  E0 → 5.1 → 5.2 → 5.3 → 5.4 → 5.5 → 5.6 → 5.7    ║
║  → 5.8 → 5.9                                      ║
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║  TRADE-OFF ENGINE (Section 3.10) [NEW]            ║
║  ATS_PRIORITY_GATE | HUMAN_PRIORITY_GATE          ║
║  SENIOR_ROLE_ADJ | FRESHER_ADJ | METRIC_STUFFING  ║
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║  AUTHENTICITY ENGINE (Section 3.9) [NEW]          ║
║  Language variation | Voice consistency           ║
║  Imperfection tolerance | Narrative coherence      ║
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L6: Output Engine → 7-10 variants
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║  VARIANT DISTANCE CONTROL (Section 3.12) [NEW]    ║
║  Structure + Signal + Pattern distance enforced   ║
║  Redundant variants removed and regenerated       ║
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                ↓
L7: Simulation + Output Validation (Section 3.4)
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║  OUTPUT PACKAGE (Section 3.7)                     ║
║  7+ variants | 7 scores per variant | Strategy Log ║
║  Sim Report | Gap Notification | ATS Advisory     ║
║  Bias Audit | Authenticity Score | Trade-Off Log  ║
╚═══════════════════════════════════════════════════╝
                ↓  [outcome reported by candidate]
╔═══════════════════════════════════════════════════╗
║  FAILURE DIAGNOSIS LAYER (Section 3.11) [NEW]     ║
║  Per-application diagnostic tree                  ║
║  Root cause → engine → actionable fix             ║
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                ↓  [50+ samples per cluster-level]
Compound Learning System (Section 3.5)
```

SOURCE: Supersedes Section 2.1 pipeline diagram. Reflects all additions in Sections 3.8–3.13.

LAYER 3

Execution Specification

How it is built and run. Full contracts. No strategy language.

3.1 VALID RUN DEFINITION

A valid Resume OS run satisfies ALL of the following. Any condition not met = invalid run.

- Input completeness: Master resume provided in parseable format. JD text OR job URL provided.
- Constraint Engine has run: hard fail conditions checked before any layer executes.
- Engine 0 (Signal Architecture) completed: Signal Classification Index produced before any other engine runs.
- Parsing Gate passed: at least one variant passes structural compliance.
- Minimum 7 variants generated: fewer than 7 = invalid run.
- Strategy Log produced: every change has a before/after entry with rationale.
- All 5 scores calculated for every variant.
- Integrity check passed: every claim in every variant traceable to master resume.

Invalid run conditions (any one triggers invalidity):

- Engine 0 skipped — downstream signal type tags are unverifiable
- Parsing Gate not run — ATS scores are not valid
- Any content added not present in master resume — FABRICATION, hard block
- Fewer than 7 variants generated
- ATS score reported without naming the detected ATS platform

`SOURCE: v1.0 Sec 10 + v2.0 Sec 11 + v3.0 Sec 4 + v3.3 Sec 1 + Patch G`

3.2 THE 9 SUB-ENGINE CONTRACTS

> *Engines run in strict sequence: 0 → 5.1 → 5.2 → 5.3 → 5.4 → 5.5 → 5.6 → 5.7 → 5.8 → 5.9. Each engine produces structured output that is the input to the next. No engine is optional. An engine with no changes still runs and logs a null-change entry in the Strategy Log.*

```
Engine 0  (Signal Architecture)   → required before all others
Engine 5.1 (Positioning)           → requires Engine 0 output
Engine 5.2 (Keyword)               → requires Engine 5.1 output
Engine 5.3 (Bullet)                → requires Engine 5.2 keyword map
Engine 5.4 (Role-Shift)            → requires Engine 5.3 output
Engine 5.5 (Environment)           → requires Engine 5.4 output
Engine 5.6 (Signal Amplification)  → requires Engine 5.5 + Engine 0 priority stack
Engine 5.7 (Recency Weighting)     → requires Engine 5.6 output
Engine 5.8 (Structure)             → requires Engine 5.7 output
Engine 5.9 (Bias Mitigation)       → requires Engine 5.8 output (near-final)
```

Engine 0: Signal Architecture Engine

Purpose: Classifies every element of the working copy into one of four signal types (Hard/Structural/Cognitive/Trust). Assigns cluster-specific priority stack. Identifies signal gaps before any rewriting. Produces the Signal Classification Index used by all subsequent engines.

Input: Working copy of master resume + Role DNA (Function + Level + Cluster + Environment)

Output: Signal Classification Index: every resume element tagged with signal type + priority rank + placement recommendation. Gap delta list: elements with zero Hard Signals, zero Trust Signals, misplaced signals.

Transformation Logic:

- Parse every bullet: extract action verb, scope, metric/outcome, any named entity. Tag: Hard (quantified metric present), Cognitive (problem→action→outcome structure present), Trust (institutional name/certification/URL present), Structural (formatting compliance only — no content tag)
- Apply cluster priority stack from Signal Architecture table (Section 2.4). Record Priority 1–4 for the target cluster.

- Flag every bullet with zero Hard Signals as 'metric-absent'. Flag every section with no Trust Signals as 'credibility-gap'. Flag any cognitive signal placed below the fold as 'placement-mismatch'.
- Output is a structured classification index. Every subsequent engine reads this index before making changes.

Sequence Dependency: Must run BEFORE Engine 5.1. No other engine may independently classify signal types.

Research Basis: FACT (Pina et al. 2023, Tobii eye-tracking n=221): Skills section shows no fixation difference between accepted/rejected. INFERENCE: Signal type determines attention probability.

Failure Modes:

- Classifying 'Responsible for X' as Cognitive — contains no problem→action→outcome structure
- Treating company names as Hard Signals — they are Trust Signals regardless of revenue associations

`SOURCE: v3.0 Sec 4.5 Engine 0 + v3.3 Sec 4 + Research Doc Domain 1`

Engine 5.1: Positioning Engine

Purpose: Rewrites the professional summary for precise role-specific positioning. Answers three questions: who you are + for whom + driving what outcome. Minimum one Hard or Trust Signal in first two lines.

Input: Engine 0 Signal Classification Index + Role DNA (Level + Function + Environment) + Existing summary

Output: Rewritten summary: level-calibrated length, role-specific framing, Hard/Trust Signal in lines 1–2. Strategy Log entry: before/after + rationale.

Transformation Logic:

- Target length by Level: L1-L2 → 3 lines | L3 → 3–4 lines | L4 → 4–5 lines | L5+ → 5–6 lines
- Check Engine 0 index: does current summary contain Hard or Trust Signal in first two lines? If no → pull highest-priority signal from index and integrate into line 1 or 2
- Apply three-question frame: Line 1 = who you are (functional identity + years + specialisation). Line 2 = for whom (Environment match) + outcome you drive (must reference real achievement). Lines 3+ = level-appropriate directional signal.
- Validate: no generic openers. No soft skill claims without evidence. Every sentence tied to target role function and level.
- If summary already passes all checks → log null-change. Do not force rewrites that degrade quality.

Sequence Dependency: Must complete before Engine 5.2, which distributes keywords into the rewritten summary.

Research Basis: FACT (Resume Genius 2024): 90% of HMs say clear summary speeds evaluation; 29% deterred by generic summaries. FACT (Willis & Todorov 2006): 100ms impression formation.

Failure Modes:

- Summary without Hard/Trust Signal in lines 1–2 — fails Triage Zone proof check
- Generic opener ('Results-driven professional') — activates Bypassing Pattern before first word is read

`SOURCE: v1.0 Sec 5.1 + v2.0 Sec 7 Sub-Engine 5.1 + v3.0 Sec 4.5`

Engine 5.2: Keyword Engine

Purpose: Maximises ATS keyword coverage without stuffing. Distributes keywords across three zones by ATS weighting. Enforces orphaned-keyword rule: no skill appears in Skills section without evidence in at least one Experience bullet.

Input: JD signal extraction (required, preferred, tool names, KPI vocabulary) + Engine 5.1 output + Working copy experience section + Engine 0 index

Output: Keyword Distribution Map: which exact keywords appear in Summary, Experience, Skills. Coverage%. Semantic variant list. Orphan flags. Updated working copy.

Transformation Logic:

- Extract from JD: Required keywords ('must have / required / essential'), Preferred ('nice to have'), exact tool names (exact JD spelling), KPI vocabulary
- Expand with semantic variants per ATS type: Greenhouse/Lever (semantic) → add ML/machine learning/AI models all three. Workday/Taleo (exact match only) → JD exact string only
- Distribute by ATS priority: Zone 1 Summary (2–4 core keywords natural) → Zone 2 Experience (keywords in bullet context with evidence) → Zone 3 Skills (full plain comma-separated list)
- Orphan check: any skill in Skills with zero evidence in any Experience bullet → flag 'ORPHANED_KEYWORD'. Skill remains for ATS but logged as human credibility risk.
- Coverage check: coverage% = (JD required keywords present ÷ total JD required) × 100. If < 70% → Constraint Engine BLOCK. Log all missing keywords as development signals.

Sequence Dependency: Runs after 5.1. Engine 5.3 needs the keyword map to embed keywords in Experience bullets.

Research Basis: FACT (Jobscan 2025, n=384): 99.7% of recruiters use keyword filters. FACT (EDLIGO 2025): 66% of ATS cannot understand synonyms — exact match remains essential.

Failure Modes:

- All keywords in Skills section only — orphaned keywords are ATS boost but human credibility risk
- Semantic variants used for Workday/Taleo — these are exact-match systems; variants not parsed as equivalents

`SOURCE: v1.0 Sec 5.2 + v2.0 Sec 7 Sub-Engine 5.2 + v3.0 Sec 4.5`

Engine 5.3: Bullet Engine

Purpose: Rewrites every Experience bullet using role-calibrated, level-calibrated, cluster-specific formula. Enforces Hard Signal minimum (1 metric per 2 bullets). Strips passive openers. Applies primacy/recency ordering within each role section.

Input: Engine 0 Signal Classification Index (every bullet pre-tagged) + Role DNA (Cluster + Level) + Engine 5.2 Keyword Map

Output: Rewritten experience bullets: action-verb-first, level-calibrated formula, minimum metric density, primacy/recency ordered. Signal type tag retained in working copy metadata.

Transformation Logic:

- For each bullet: read Engine 0 tag. Hard Signal → cluster-specific Hard formula. Cognitive → problem→action→outcome restructure. Structural/Trust → passive-opener check only.
- Master formula: [Action Verb (ownership-matched to Level)] + [Scope: what/who/scale] + [Impact: business outcome] + [Metric: quantified result]. Impact before task. Outcome before method.
- Passive opener check: 'Responsible for', 'Worked on', 'Helped', 'Participated', 'Assisted', 'Supported', 'Was involved in' → replace with hierarchy: Architected > Designed > Led > Built > Shipped > Delivered > Managed > Contributed
- Metric density: if first 2 bullets of any role have zero quantified metrics → flag. Apply scope proxy (team size, volume, frequency) if outcome metric unavailable.
- Ordering: Bullet #1 = strongest Hard Signal (primacy, Murdock 1962). Bullet #last = secondary strength (recency, Glanzer & Cunitz 1966). Middle = supporting detail.
- Bold rule: max 3 bold elements per bullet. Bold priority: metric value > Trust Signal name > outcome phrase.
- Length: 1–2 lines per bullet. >2 lines → split or cut to core claim.

Sequence Dependency: Runs after 5.2 (keyword map needed). Engine 5.4 reads Bullet Engine output to adjust level language.

Research Basis: FACT (TalentWorks, n=6,305): quantified achievements = ~40% higher interview conversion. FACT (Cultivated Culture, n=125,000+): 36% zero metrics. FACT (NNG Spotted Pattern): numerals and bold text

attract disproportionate pre-attentive fixation. FACT (Murdock 1962, Glanzer & Cunitz 1966): serial position effects. FACT (Cowan 2001): 4±1 chunk limit.

Failure Modes:

- Strongest achievement in bullet #3 — violates primacy rule; recruiter likely skips during triage
- Spelling out numbers ('thirty-four percent') — must be numeric ('34%') to trigger Spotted Pattern fixation

`SOURCE: v1.0 Sec 5.3 + v2.0 Sec 7 Sub-Engine 5.3 + v3.0 Sec 4.3 + Research Docs 1, 3, 4`

Engine 5.4: Role-Shift Engine

Purpose: Scans all content for language signalling a level lower than the target. Rewrites to level-appropriate language without changing underlying facts. Prevents the core failure mode: seniority mismatch between resume language and target role.

Input: Engine 5.3 output + Role DNA Level axis + Level Language Rules (Section 2.2)

Output: Level-corrected working copy. Strategy Log flagging every shift: before/after. Warning if structural level gap exists (evidence genuinely does not support target level — language shift cannot resolve this).

Transformation Logic:

- Scan all bullets for level-signal mismatch: compare action verb strength against target level expectation.
- Apply shift logic: L1→L3: 'I helped build' → 'Owned… designed… distributed to N stakeholders'. L2→L4: 'Worked on X' → 'Led X decomposition; designed architecture; managed N-engineer pod'. L3→L5: 'Led a team' → 'Scaled team from N to N; defined hiring criteria; established engineering standards'.
- Distinguish language shift from fabrication: if master resume contains ownership evidence (verbs: designed, defined, led, owned) but current language is passive → shift is valid. If master resume shows only participation → shift cannot be made. Flag structural gap in Strategy Log.
- Reverse check: mid-level candidate claiming L4 language without evidence → flag as risk signal, do not inflate.

Sequence Dependency: Runs after 5.3. Engine 5.5 adjusts tone — 5.4 must correct level language first.

Research Basis: INFERENCE (Derous & Ryan 2019 + Interviewing.io 2024): Level signals in resume language are heuristically processed through verb choice before content evaluation begins.

Failure Modes:

- Applying L4 language to L1-L2 evidence — level inflation is a fabrication variant, not an optimisation
- Not shifting when master resume DOES contain ownership evidence but writing uses passive voice — missed optimisation

`SOURCE: v1.0 Sec 5.4 + v2.0 Sec 7 Sub-Engine 5.4 + v3.0 Sec 4.5`

Engine 5.5: Environment Adaptation Engine

Purpose: Adjusts tone and signal framing to match the cultural hiring bias of the target organisation type. Same facts — different emphasis. Does not change what the candidate did; changes how it is framed relative to employer values.

Input: Engine 5.4 output + Role DNA Environment axis + Amplify/Suppress language pairs

Output: Environment-adapted working copy. No new content — framing adjustments only.

Transformation Logic:

- Startup (0–200 emp) — Amplify: 'built from scratch', 'owned end-to-end', 'zero to one', 'shipped in X weeks', 'operated without playbook'. Suppress: governance, approval chains, committee decisions.
- SME (200–2,000) — Amplify: 'cross-functional', 'wore multiple hats', 'delivered across X and Y simultaneously', 'pragmatic solutions'. Balance specialist depth with versatility.
- MNC (2,000+) — Amplify: 'managed across N regions', 'compliant with X framework', 'scaled within structured programme'. Suppress: ambiguity tolerance language, solo execution framing, no-process signals.

- Apply to Summary first, then Experience bullets. If amplify word doesn't fit the achievement structure naturally → do not inject artificially.

Sequence Dependency: Runs after 5.4. Engine 5.6 reads environment-adapted content to amplify highest-value signals.

Research Basis: INFERENCE (Derous & Ryan 2019): Hiring bias by organisation type is well-documented. ASSUMPTION: Environment framing adjustment → higher interview conversion (pending outcome data validation).

Failure Modes:

- Startup framing on an MNC application — signals culture misfit pre-attentively
- Injecting environment language that doesn't match the achievement — artificial framing creates cognitive dissonance

`SOURCE: v1.0 Sec 5.5 + v2.0 Sec 7 Sub-Engine 5.5 + v3.0 Sec 5.2`

Engine 5.6: Signal Amplification Engine

Purpose: Surfaces the two highest-priority signal types per role cluster (from Engine 0 priority stack) and increases their density and prominence in the triage zone. Applies leadership verb hierarchy. Adds scale indicators where outcome metrics unavailable.

Input: Engine 5.5 output + Engine 0 Signal Classification Index (priority stack) + Role DNA (Cluster)

Output: Signal-amplified working copy: Priority 1 and Priority 2 signals prominent in triage zone. Leadership verb hierarchy applied. Scale indicators added where missing.

Transformation Logic:

- Read Engine 0 priority stack for target cluster. Identify Priority 1 and Priority 2 signal types.
- Check: are Priority 1 signals present in triage zone (top 40%, specifically Summary and first 2 bullets of most recent role)? If no → move strongest Priority 1 signal from later position into triage zone. Log relocation in Strategy Log.
- Apply leadership verb hierarchy to all action verbs: Architected > Designed > Led > Built > Shipped > Delivered > Managed > Contributed. Never apply a verb the candidate's evidence does not support.
- Scale indicator addition: for any bullet with no quantified metric, add strongest available scope proxy: team size, user count, transaction volume, budget managed, geographic scope. Log as 'scope proxy applied'.
- Bold enforcement: max 3 bold elements per bullet. Priority: metric value > Trust Signal name > outcome phrase.

Sequence Dependency: Runs after 5.5. Engine 5.7 compresses bullet density — 5.6 must amplify signals first so correct content is preserved through compression.

Research Basis: FACT (Interviewing.io 2024): Trust signals produce 35% interview rate lift. FACT (NNG Spotted Pattern): Numerals and bold attract disproportionate pre-attentive fixation. INFERENCE: Moving Priority 1 signal into triage zone directly increases Commitment activation probability.

Failure Modes:

- Amplifying non-Priority-1/2 signals — wastes triage zone real estate
- Applying 'Architected' when evidence only supports 'Contributed' — verb inflation is a fabrication variant

`SOURCE: v1.0 Sec 5.6 + v2.0 Sec 7 Sub-Engine 5.6 + v3.0 Sec 4.5 + Signal Priority Stack (Section 2.4)`

Engine 5.7: Recency Weighting Engine

Purpose: Applies bullet density rules by role age. Compresses older experience progressively to prevent cognitive overload and concentrate recruiter attention on recent relevant experience.

Input: Engine 5.6 output + Role DNA (Level) + Role recency map

Output: Recency-weighted working copy: current role maximum density, older roles compressed. Exception flagged when older role is more JD-relevant than current role.

Transformation Logic:

- Map roles to recency bands: Band 1 (current) → Band 2 (1 prior) → Band 3 (2 prior) → Band 4 (3+ prior)
- Compression rules: Band 1: 4–6 bullets, full formula. Band 2: 3–4 bullets. Band 3: 2–3 bullets, core metrics only. Band 4: 1–2 bullets OR title + company + dates only.
- Education: at 7+ years → 2 lines only (degree + institution + year). No coursework, no GPA unless employer known to filter AND GPA ≥ 8.0.
- Exception check: if older role has higher keyword overlap with target JD than current role → log in Strategy Log: 'Older role [X] has higher JD relevance. Recommend variant that surfaces this role prominently.'
- v3.3 Hard Fail: if any role exceeds 5 bullets → Constraint Engine triggers HARD FAIL. Compress further.

Sequence Dependency: Runs after 5.6. Engine 5.8 sets final section order — 5.7 must compress content first so Structure Engine knows total volume per section.

Research Basis: FACT (TheLadders 2018): Dates in the 6 critical triage fixation points — older roles receive progressively less attention. FACT (Cowan 2001): 4±1 working memory chunks.

Failure Modes:

- Compressing current role to 3 bullets to fit one page — violates Band 1 minimum; signals low impact
- Equal bullet density across all roles — maximises cognitive load, dilutes signal from current role

`SOURCE: v1.0 Sec 5.7 + v2.0 Sec 7 Sub-Engine 5.7 + v3.0 Sec 4.3`

Engine 5.8: Structure Engine

Purpose: Applies section ordering, header naming, and physical layout rules based on cluster + level + ATS combination. Validates the pre-attentive design layer. Enforces all ATS parsing constraints. Finalises the complete architecture of the resume.

Input: Engine 5.7 output + Role DNA (Cluster + Level) + ATS Intelligence output (platform, file format) + Pre-Attentive Layer rules

Output: Fully structured working copy with section order, headers, layout finalised. Parsing Gate validation result (PASS/FAIL). Pre-attentive quality flags.

Transformation Logic:

- Apply section ordering for cluster + level combination. Universal: standard section headers only ('Work Experience' not 'Where I've Been').
- ATS Parsing Gate: (a) single-column for Workday/Taleo/SAP — multi-column = FAIL. (b) No tables in body = FAIL. (c) No icons/rating bars = FAIL. (d) Contact block in document body NOT Word header/footer. (e) Correct file format for detected ATS. Any failure → STOP.
- Pre-Attentive Layer validation: whitespace ratio ≥20% per page. Interline spacing 1.15–1.3. Three typographic levels maximum. Hard left-alignment all body content. Calibri or Tier-1 font. One accent colour maximum — none in body text.
- Contact block audit: Name (largest text on page) + Location + Phone + Email + LinkedIn URL + GitHub/Portfolio URL. Missing LinkedIn → flag. Missing GitHub for C1/C3/C5 → flag.
- Section header standardisation: 'My Journey' → 'Work Experience'. 'Tech Stack' → 'Skills'. Non-standard headers cause ATS misclassification.

Sequence Dependency: Penultimate engine before 5.9 (Bias Mitigation). Output is near-final working copy.

Research Basis: FACT (EDLIGO 2025, n=1,000): 43% of ATS rejections are non-qualification-related — 23% parsing errors, 12% formatting. FACT (TopResume): 25% of ATS systems cannot read Word header/footer content. FACT (NNG): Users spend 80% of viewing time on left half of page.

Failure Modes:

- Creative section headers — ATS misclassifies; section content is lost

- Contact info in Word header element — invisible to 25% of ATS systems
- PDF for Workday — 18% failure rate vs 4% for DOCX

SOURCE: v1.0 Sec 5.8 + v2.0 Sec 7 Sub-Engine 5.8 + v3.0 Sec 4.6 + v3.3 Sec 2

Engine 5.9: Bias Mitigation Engine

Purpose: Reviews near-final working copy for patterns known to trigger demographic bias in human and AI screening. Applies three targeted interventions. Does not change factual content — only documented bias-trigger language and formatting patterns.

Input: Engine 5.8 output + Bias pattern library (gender language, gap formatting, institution-emphasis rules)

Output: Bias-audited final working copy. Per-variant bias audit flag in scoring package. Strategy Log entry for each intervention.

Transformation Logic:

- Gender-neutral language scan: replace gendered pronouns and adjectives with role-descriptive language. (Resumly: gendered wording reduces callback rates for women up to 25%.)
- Employment gap date formatting: gaps ≥ 6 months → flag 'years-of-experience format' as option. CANDIDATE CONSENT REQUIRED — log as recommendation, not automatic change. (Harvard Kennedy School RCT, n=9,022: 15% callback lift.)
- Assessment-forward placement for freshers: if candidate is L1-L2 with PickCV assessment scores → verify score is in header block adjacent to LinkedIn URL. If buried in Education → relocate. (Deloitte 2024: 93% campus recruiters weight technical interview performance as pivotal.)
- Institution-name weight for freshers: if institution name formatted more prominently than project outcomes or assessment scores → adjust. Trust hierarchy for freshers: verified assessment > live portfolio > institution.
- Bias audit flag: per variant, flag which bias dimensions addressed. Required for platforms using Resume OS in AI-augmented hiring pipelines. (UW 2024: LLM hiring mirrors demographic bias shown to it.)

Sequence Dependency: Final engine. Output of 5.9 feeds Layer 6 (Output Engine) as candidate-ready working copy.

Research Basis: FACT (UW 2024): AI-assisted screening favoured white-associated names 85.1% of cases. FACT (Harvard Kennedy School RCT, n=9,022 + 2,650): 15% callback lift from years-of-experience format. FACT (Resumly): gendered wording reduces callbacks up to 25%. FACT (Deloitte India 2024): 93% campus recruiters weight technical interview performance.

Failure Modes:

- Auto-applying gap format without candidate consent — changes how candidate presents their history; opt-in required
- Suppressing institution names entirely — the intervention is about proportional weight, not removal

SOURCE: v3.0 Sec 4.5 Engine 5.9 + Research Doc Domain 1 (bias section) + Domain 6 (campus) + Patch A

3.3 PRE-ATTENTIVE LAYER — FORMAL RULES WITH THRESHOLDS

> *All rules enforced by Engine 5.8 (Structure Engine). Thresholds are measurable, not subjective. WITHDRAWN CLAIM: '20% comprehension gain from whitespace' (Galitz 2007 citing Lin 2004) has been debunked. Carl Myhill confirmed Lin 2004 paper contains no such finding. The whitespace benefit is real (Chaparro et al. 2004, Wichita State, p=.01) but without a percentage claim.*

Rule	Tag	Measurable Threshold	Enforcement Logic	Failure Mode
WHITESPACE_RATIO	FACT	≥20% of page area empty (whitespace	Engine 5.8: if <20% on any page → Recency Weighting Engine	Dense layout → negative pre-attentive competence judgment in 100ms

		ratio). Measurement: (page area - text area) ÷ page area × 100	compresses lower-priority content. If compression insufficient without core content loss → generate two-page variant. One-page constraint does NOT override whitespace minimum.	window. Cannot be recovered during triage.
SECTION_SPACING	INFERENCE	Consistent vertical spacing between all sections. Variation across section gaps: ≤2pt difference	Engine 5.8: uniform before-spacing on all section header paragraphs. Recommended: 240–320pt before, 60–80pt after. Consistency is the requirement, not specific value.	Inconsistent rhythm breaks pre-attentive processing — forces conscious attention to layout instead of content.
INTERLINE_SPACING	FACT	1.15–1.3× font size for print/DOCX. 1.5× minimum for accessibility (WCAG 2.2)	Engine 5.8: sets line spacing value. ATS format (DOCX): 1.15 sufficient. Human-read (PDF): 1.3 preferred.	0.8× spacing significantly impairs comprehension (Rello & Pielot 2016).
HIERARCHY_LEVELS	INFERENCE	Maximum 3 distinct typographic levels (size + weight). Level 0 = Name (contact header only)	Engine 5.8: enforces 3 levels. Level 1 = Section Headers (largest, bold). Level 2 = Role/Company (medium). Level 3 = Body text (base). Size difference between adjacent levels: 2–6pt.	>3 levels creates visual noise → Layer-Cake pattern cannot activate → F-pattern fallback.
FONT_COMPLIANCE	FACT	All fonts in Tier-1 ATS-safe list: Calibri (default), Arial, Helvetica, Verdana, Georgia, Cambria, Times New Roman	Engine 5.8: replace non-compliant fonts with Calibri. No script, decorative, or display fonts.	Non-standard fonts reduce ATS text extraction accuracy. Some fonts render as garbled characters in older parsers.
LEFT_MARGIN_ALIGNMENT	FACT	All body text hard-left-aligned to document left margin. Zero indentation variation except bullet levels.	Engine 5.8: hard left alignment enforced. Contact Name may be centred. Bullet indent levels permitted (440 DXA left, 280 DXA hanging).	Inconsistent left alignment disrupts F-pattern and I-pattern navigation spine — recruiter misses left-anchored signal.
COLOUR_CONSTRAINT	FACT	Maximum one accent colour. Zero colour in body text, bullets, or skills section.	Engine 5.8: one accent colour only (section headers or name). All body text #1F2937 or equivalent dark. Flag any coloured body text.	Coloured body text reduces ATS keyword extraction accuracy (EDLIGO 2025). Colour in body competes with Spotted Pattern triggers.

Cognitive Load Budget — Measurable Parameters

Parameter	Threshold	Measurement	Enforcement Engine	Research Basis
Bullets per role — current	4–6. Hard Fail if >5 (v3.3)	Count bullet items under most recent role header	Engine 5.7 (Recency), then Constraint Engine Hard Fail check	Cowan 2001: 4±1 working memory chunks
Bullets per role — Band 2	3–4	Count under second-most-recent role	Engine 5.7	TheLadders 2018: older roles receive progressively less attention
Bullets per role — Band 3	2–3	Count under third-most-recent role	Engine 5.7	TheLadders 2018 + Cowan 2001

Bullets per role — Band 4+	1–2 or title + company + dates only	Count under all roles beyond Band 3	Engine 5.7	TheLadders 2018
Words per bullet	15–30 optimal. Flag at >35. Max 40.	Word count of each bullet after Engine 5.3 rewrite	Engine 5.3 flags; recommends split or cut	Cowan 2001 chunk limit + WCAG readability
Sections per page 1	Maximum 6 top-level section headers above fold	Count section headers on page 1	Engine 5.8: if >6 → compress or merge non-critical sections	INFERENCE: >6 sections breaks Layer-Cake navigation efficiency
Serial position adherence	Bullet #1 = strongest Hard Signal. Bullet #last = second strength.	Engine 5.3 validates ordering post-rewrite	Bullet Engine enforces; Strategy Log notes if ordering changed from master resume	Murdock 1962 + Glanzer & Cunitz 1966
Bold element density	Maximum 3 bold elements per bullet	Count bold runs per bullet after Engine 5.6	Engine 5.6 enforces; removes excess bold from lowest-priority elements	NNG Spotted Pattern: desensitisation above 3 bold elements per visual unit

SOURCE: v2.0 Sec 2 (Pre-Attentive Design Layer) + v3.0 Sec 4.6 + Research Doc Domain 4 + Cowan 2001 + Murdock 1962 + Rello & Pielot 2016

3.4 RECRUITER DECISION SIMULATION LAYER [NEW — VFINAL UNIFIED]

> *This layer addresses the gap identified in the system critique: 'You score resumes but you don't simulate recruiter decisions beyond heuristics.' The Simulation Layer does not replace scoring — it supplements it with a decision probability model grounded in the documented accuracy and bias patterns of human recruiters.*

3.4.1 Why Simulation Is Needed

The current scoring system (ATS + Recruiter Scan + Signal Density + Cognitive Load + Role Match) measures resume quality in isolation. It does not model the recruiter's actual decision process, which is:

- 55% accurate at predicting candidate interview performance (Interviewing.io 2024) — barely above a coin flip
- Dominated by pedigree heuristics: FAANG experience = 35% interview rate lift, irrespective of stated selection criteria
- Affected by cognitive load: high-load documents force heuristic (System 1) mode, reducing accuracy further
- Affected by confirmation bias: first visible signal anchors all subsequent interpretation (Rabin & Schrag 1999, Asch 1946)

The simulation layer models THESE factors, not just document quality. A resume with a 90% Signal Density Score but no Trust Signal in the Primary Optical Area may still get rejected by a recruiter operating in heuristic mode. The simulation surfaces this risk.

3.4.2 The Five Simulation Dimensions

Dimension	What Is Modelled	Input	Output	Research Basis
Triage Pass Probability	Probability that the document passes the 6-data-point triage check	Position + visual weight of: Name, Current	Score 0–100%: probability all 6 data points are found within the 7–31 second	FACT: TheLadders 2018 — 6 critical fixation anchors. 80% of

	based on where the six critical elements appear and their visual prominence.	Title/Company, Previous Title/Company, Current Dates, Previous Dates, Education in the working copy output.	triage window. Flag any data point not in Triage Zone (top 40%).	triage attention on these 6 elements.
Pedigree Heuristic Score	Presence and prominence of institutional Trust Signals that activate the pedigree heuristic in recruiter decision-making.	List of company names and institutions in working copy. Position in document (Primary Optical Area vs below fold).	Score 0–100%: weighted by number of recognisable Trust Signals × positional premium (above fold = 2× weight). NOT correlated with ATS score — recruiter heuristic, not algorithm.	FACT: Interviewing.io 2024 — FAANG experience = 35% interview rate lift, single strongest predictor. Trumps stated rejection reasons.
Confirmation Bias Anchor	Whether the first substantive content element encountered by the recruiter creates a positive or neutral interpretive frame for everything below.	First non-contact element visible in Triage Zone: is it a Hard Signal (positive), Trust Signal (positive), a risk signal (negative), or a neutral element?	Flag: POSITIVE ANCHOR (Hard or Trust Signal first) | NEUTRAL ANCHOR | NEGATIVE ANCHOR (risk signal first). Negative anchor = all subsequent content interpreted through negative frame (Rabin & Schrag 1999).	FACT: Rabin & Schrag 1999 (QJE): people misinterpret confirming new information for prior hypotheses. FACT: Asch 1946 (10 studies): first-seen traits anchor all subsequent impression formation.
Cognitive Load Decision Mode Prediction	Whether the document's cognitive load level is likely to keep the recruiter in analytical (System 2) or heuristic (System 1) mode during the evaluation read.	Cognitive Load Score (from scoring system Section 2.7) + Triage Pass Probability.	Prediction: SYSTEM 2 LIKELY (Cognitive Load Score ≥70% + Triage Pass ≥80%) | MIXED (either condition not met) | SYSTEM 1 LIKELY (Cognitive Load <50% or Triage Pass <60%). System 1 mode = recruiter defaults to pedigree heuristic; 55% accuracy ceiling applies.	FACT: Cowan 2001 (cognitive load). INFERENCE: Kahneman dual-process model (System 1/2). INFERENCE: High cognitive load → heuristic mode → lower decision quality.
Risk Signal Detection	Count and severity of risk signals that increase recruiter's perceived probability of a bad hire.	Working copy full text scanned for risk signal list: vague ownership, no metrics, inconsistent trajectory, over-generalisation, 'responsible for' language, soft skill claims without evidence, 'familiar with' language, college projects at 5+ years experience.	Count of hard risks (each = 10-point penalty on Recruiter Scan Score). Count of soft risks (each = 3-point penalty). HARD RISKS: any single one of the list above. SOFT RISK: 2+ instances of the same pattern.	INFERENCE (v1.0 Sec 4.2 Risk Signal Register): Hiring is a risk decision. Every risk signal increases the recruiter's probability of a pass. Research basis: Rabin & Schrag 1999 (confirmation bias amplifies negative signals once noticed).

3.4.3 Simulation Score Calculation

The Simulation Report is produced for each variant alongside the existing 5 scores. It is NOT a replacement for the Recruiter Scan Score — it models the human decision layer that the Recruiter Scan Score approximates but does not fully capture.

```
Simulation Score (0-100%) =
  Triage Pass Probability         × 0.35
+ Pedigree Heuristic Score        × 0.30
+ Confirmation Bias Anchor        × 0.20  (POSITIVE=100, NEUTRAL=60, NEGATIVE=10)
+ Cognitive Load Decision Mode    × 0.15  (SYSTEM2=100, MIXED=60, SYSTEM1=20)

Risk Signal Penalty (applied after):
  Each HARD risk signal  → -10 points
  Each SOFT risk signal  → -3 points
  Floor: 0 (cannot go negative)

Interpretation thresholds:
  85-100%: High probability of recruiter Commitment activation
  70-84%:  Likely triage pass; evaluation read quality depends on content
  55-69%:  Triage pass uncertain; pedigree signal absent or weak
  <55%:    Low probability of Commitment; review Pedigree Score + Anchor
```

> *The Simulation Score is tagged INFERENCE. It is a model derived from documented recruiter behaviour patterns — it is not empirically validated against actual Resume OS application outcomes. It becomes FACT-grounded once 50+ applications per cluster-level provide outcome data. Until then, treat it as a diagnostic tool, not a decision gate.*

3.4.4 Output Validation — What the System Must Check Before Delivery

Beyond the Simulation Score, every Resume OS output must pass a pre-delivery validation checklist. This is the system's self-check before any variant is delivered to the candidate.

Check	What Is Validated	Pass Condition	Fail Action
Fabrication audit	Every claim, metric, and credential in every variant traceable to master resume	100% traceability — zero unsourced content	HARD BLOCK: remove fabricated content, re-run affected engines, re-generate variant
Level consistency	Resume language level consistent with master resume evidence (not just target level)	No verb or scope descriptor beyond what master resume evidence supports	HARD BLOCK: revert to evidence-supported language, flag in Strategy Log
Parsing Gate	All ATS structural constraints met for the detected platform	PASS on all platform-specific constraints	HARD BLOCK: do not deliver INVALID-tagged variants
Variant minimum	At least 7 distinct structural variants generated	7 or more variants with different structure/emphasis	BLOCK run: invalid until minimum met
Score completeness	All 5 scores + Simulation Score calculated for every variant	No variant with missing scores	BLOCK delivery: recalculate missing scores
Strategy Log completeness	Every change in every variant has a Strategy Log entry	Before/after + rationale for every delta from master resume	BLOCK delivery: complete log before delivering
Bias audit flag	Engine 5.9 has produced a bias audit flag for every variant	Boolean per dimension (gender-neutral, gap-format, assessment-forward) per variant	FLAG but do not block: note missing audit in ATS Advisory
Application timing signal	ATS Advisory includes posting age and timing recommendation if posting date available	Advisory notes whether within 96-hour optimal window	NOTE in advisory if posting date unavailable

SOURCE: New in vFinal Unified: Simulation Layer + Output Validation. Grounded in: Interviewing.io 2024 (recruiter accuracy) + TheLadders 2018 (triage anchors) + Rabin & Schrag 1999 (confirmation bias) + Asch 1946 (impression formation) + Cowan 2001 (cognitive load) + Kahneman dual-process model

3.5 CONTINUOUS LEARNING — EXECUTION SPECIFICATION

The learning system activates only when the minimum sample threshold is reached. Before threshold, rules are ASSUMPTION-tagged design hypotheses. After threshold, rules are FACT-grounded system defaults.

```
Per application logged:
  Role DNA (Function + Level + Environment + KPIs)
  ATS detected + formatting rules applied
  Gap Scorecard (all 5 dimensions)
  Variant submitted (which of V1-V10)
  Strategy Log delta (all changes made)
  Simulation Score + 5 scores at time of delivery
  Outcome (candidate-reported): No response | Recruiter screen | Technical |
                                Final round | Offer | Decline
  Time to first recruiter contact (if tracked)

Pattern analysis trigger condition:
  Minimum 50 applications in same [Cluster × Level] combination
  Before 50 samples: rules are ASSUMPTION
  After 50 samples: pattern conclusions may modify system defaults

What CAN be modified by outcome data:
  Default recommended variant per cluster (V1-V10 → which produces highest response rate)
  Keyword priority weighting
  Summary length optimality by experience band
  Bullet density sweet spot per cluster-level
  Simulation Score weights (calibrate against actual recruiter decisions)

What CANNOT be modified by any learning data:
  FABRICATION BLOCK — never overridden
  LEVEL MISREPRESENTATION — never overridden
  ATS PARSING GATE hard constraints — never overridden
  CANDIDATE REVIEW requirement — never bypassed
  50-SAMPLE MINIMUM — never lowered
```

SOURCE: v1.0 Sec 9 + v2.0 Sec 9 + v3.0 Sec 9 + Patch G

3.6 OPEN ITEMS — GAPS NOT YET RESOLVED

> *These are documented gaps, not omissions. They are listed here so the system is not treated as complete in areas that remain unspecified.*

Gap ID	Area	What Is Missing	Impact	Status
GAP-01	University DNA Configuration	Pick Camp product spec references 'University DNA configuration'. No implementation spec exists — what it contains, how captured, how it modifies Resume OS behaviour.	High — core B2B differentiator without an implementation spec	OPEN — requires product definition
GAP-02	Cover Letter Scope	Engine 5.9 references cover letters as a Cognitive Signal placement zone. Resume OS has no cover letter generation spec, scope statement, or scoring model.	Medium — creates ambiguity about system boundary	OPEN — either scope cover letters or remove all references
GAP-03	Persona Match (Pick Hire)	Pick Hire's ranking formula uses Persona Match (20%). No equivalent exists in	High — Pick Hire ranking is	OPEN — requires definition in Pick Hire spec

		Resume OS Signal Architecture. Cannot be implemented.	incomplete without this definition	
GAP-04	Recruiter Feedback Loop	Learning system references 'recruiter feedback at screen stage' as an input type. No mechanism for collecting or routing this feedback is specified.	Medium — learning system has an unimplemented input path	OPEN — requires product feature
GAP-05	LLM Parser Optimisation	System is LLM-parser-aware (Textkernel 2023, Resume2Vec 2025) but has no formatting or content rules specific to LLM parsers beyond ESCO semantic expansion.	Medium — will become critical as LLM parsers proliferate in India	OPEN — requires research into LLM parser failure modes
GAP-06	Emerging Role Variants	7 AI-adjacent role variants (v1.0 Sec 2.5) exist in one version only and are not wired into the v3.0 Decision Engine IF/THEN rules.	Low — fall back to parent cluster; adjustments exist but not formalised	OPEN — add to Decision Engine in next iteration
GAP-07	Simulation Score Validation	Simulation Layer (Section 3.4) is tagged INFERENCE throughout. Weights (0.35/0.30/0.20/0.15) are theoretically derived. Require outcome data to calibrate.	Medium — simulation is diagnostic until validated by application data	OPEN — validate via learning system after 50+ samples per cluster-level

SOURCE: Consolidated from vFinal Sec 10 (Conflict & Gap Register) + Patch F + new GAP-07 (Simulation Layer)

3.7 COMPLETE OUTPUT PACKAGE SPECIFICATION

Output	Format	Required Contents	Gate
Resume variants	7–10 files (PDF and/or DOCX per ATS)	Complete resume in correct format and structure per variant template (V1–V10)	REQUIRED — block if <7
Score table	Structured table	Per variant: ATS% (with platform name) + Recruiter Scan% + Signal Density% + Cognitive Load% + Role Match% + Simulation%	REQUIRED
Default recommendation	Text + rationale	Single best variant. Algorithm: highest Composite Score = (ATS × 0.60) + (Recruiter Scan × 0.40). Tie-break: higher ATS.	REQUIRED
Strategy Log	Structured doc	INPUT SUMMARY → CHANGES MADE (before/after, rationale) → KEYWORD GAPS → VARIANT RECOMMENDATION	REQUIRED
Simulation Report	Text per variant	5 simulation dimensions + Simulation Score + interpretation + risk signal list with penalties applied	REQUIRED (new)
Gap notification	List	JD required + preferred keywords absent from master resume. Framed as development signals.	REQUIRED
ATS Advisory	Short text	Detected ATS platform + specific rules applied + file format recommendation + posting age + timing signal (if available)	REQUIRED
Bias Audit Flag	Boolean per dimension	Per variant: gender-neutral check + gap-format option + assessment-forward placement confirmed	REQUIRED

SOURCE: v1.0 Sec 10.3 + v2.0 Sec 8.5 + v3.0 Sec 4 execution protocol + Patch G + Section 3.4.4 (new)

3.8 INPUT NORMALIZATION ENGINE [NEW]

> *This layer runs BEFORE Engine 0. It is the system's intake gate. Without it, Engine 0 classification operates on dirty data — broken bullet structure, missing section labels, JD noise. Garbage in → garbage out is a system-level failure, not a content failure.*

3.8.1 The Problem with Raw Inputs

The system assumes structured inputs. Reality breaks this assumption in three recurring ways:

- Resume inputs: PDFs with broken parsing (multi-column tables render as garbled text), missing section headers ('Where I've Been' not mapped to 'Experience'), dates in inconsistent formats (Jan 2022, 01/22, 2022-01), metrics buried inside prose paragraphs rather than bullets, no line breaks between roles.
- JD inputs: JDs written with no distinction between required and preferred skills, responsibilities mixed with qualifications, salary descriptions interspersed with skill requirements, role-level language absent or contradictory (title says 'Senior' but requirements say '1+ years').
- Signal inputs: Metrics present but untagged (27 mentioned in a prose sentence, not as a Hard Signal metric), proxy signals like team size or geographic scope not recognised as scope indicators, verified credentials not distinguished from self-reported claims.

Without normalization, Engine 0's Signal Classification Index is built on ambiguous source material. Every downstream engine inherits that ambiguity.

`SOURCE: EDLIGO 2025: 23% of ATS failures are parsing errors. FACT: DOCX plain text has 4% failure vs 18% for PDF with embedded fonts. Normalization is a prerequisite for reliable Engine 0 output.`

3.8.2 Input Normalization Functions

Function	Input	Output	Rules	Failure Mode if Skipped
RESUME_PARSE_CLEANUP	Raw resume text (any format)	Structured resume: labelled sections, normalised date format, bullets extracted from prose paragraphs, roles clearly delimited	1. Map non-standard section headers to canonical labels: 'Where I've Been' → 'Work Experience'. 2. Normalise dates to YYYY-MM format internally. 3. Identify bullets vs prose blocks by line length + punctuation pattern. 4. Flag any block that cannot be confidently assigned to a section as [UNCLASSIFIED — needs candidate review].	Engine 0 classifies unstructured prose as body content, not bullets — misses metric signals inside paragraphs. Section misclassification causes wrong ATS section scoring.
JD_STRUCTURING	Raw JD text (URL-fetched or pasted)	Structured JD: Required skills list, Preferred skills list, Inferred skills list, Level signals, KPI signals, Environment signals, Action verbs	1. Parse sentence-level patterns: 'required/must have/essential' → Required. 'preferred/nice to have/bonus' → Preferred. Present in JD but in responsibilities section → Inferred. 2. Extract level signals: explicit years-of-experience + title seniority + reporting line. 3. Extract KPI vocabulary: any metric noun (quota, uptime, CAC, TTF, ARR, NPS) → KPI signal. 4. Flag JDs with <5 required skills as LOW_SIGNAL_JD — warn candidate that keyword coverage targets will be harder to hit.	Keyword Engine treats all JD text as equally weighted. Over-indexes on preferred skills. Misses inferred skill signals. Keyword coverage% calculation is inaccurate.
METRIC_INFERENCE_TAGGING	Extracted bullets from resume (post-RESUME_PARSE_CLEANUP)	Each bullet tagged: EXPLICIT_METRIC (contains number), PROXY_METRIC	1. EXPLICIT_METRIC: bullet contains a digit or percentage or currency symbol (₹, $, €, £) — tag confirmed. 2. PROXY_METRIC: bullet contains words in proxy list ('team of', 'across N', 'managing', 'covering', 'serving',	Engine 5.3 Bullet Engine cannot accurately enforce metric density rules if it doesn't know which bullets already have signals. Without tagging, scope proxies are

		(contains scope indicator: team size, geographic scope, volume word), MISSING_METRIC (contains neither)	'supporting N users/clients/stakeholders') — tag as proxy, extract scope noun. 3. MISSING_METRIC: neither condition met — tag as missing, flag for Bullet Engine forced rewrite. Threshold: if >40% of bullets are MISSING_METRIC → log as SIGNAL_POVERTY warning in Strategy Log.	rewritten away, and metrics in unusual positions are missed.
NOISE_REMOVAL	Extracted resume text (all sections)	Cleaned resume with noise terms flagged for removal or replacement	Noise categories: (1) Unverifiable soft skill claims without evidence: 'excellent communicator', 'team player', 'detail-oriented', 'passionate about' — flag for removal or replacement with evidence. (2) Cliché openers: 'results-driven', 'dynamic', 'self-starter', 'seasoned' — flag. (3) Filler phrases: 'responsible for ensuring that', 'worked closely with the team to' — flag for compression. (4) Redundant repetition: same technology or skill mentioned in identical phrasing across 3+ bullets — flag for consolidation. Removal threshold: do NOT auto-remove — flag only. Candidate reviews all noise flags in Strategy Log.	Noise consumes Cognitive Load Budget without generating signal. Cowan 2001: 4±1 chunk limit — filler language occupies chunks that should be used for Hard Signals.
CONSISTENCY_CHECK	Structured resume post-cleanup	Consistency flags: date gaps, title trajectory anomalies, tenure patterns, format inconsistencies	1. Date gap detection: if gap between roles > 6 months → flag 'EMPLOYMENT_GAP [dates]' — route to Engine 5.9 Bias Mitigation for gap-neutral formatting option. 2. Title trajectory check: if title seniority decreases between consecutive roles → flag 'TITLE_REGRESSION [role X to role Y]' — route to Engine 5.4 Role-Shift for contextualisation strategy. 3. Tenure pattern: if candidate has 4+ roles with <12 months tenure → flag 'SHORT_TENURE_PATTERN' — high risk signal per Risk Signal Register. 4. Format inconsistency: date format varies, bullet style varies, capitalisation inconsistent → normalise all to canonical format.	Risk signals from Consistency Check feed the Recruiter Decision Model (Layer 3). Without Consistency Check, risk signals are only detected during Engine 5.9 — too late in the sequence.

3.8.3 Normalization Output Contract

The Input Normalization Engine produces a Normalized Input Package that replaces the raw master resume as the input to Engine 0. The master resume is never modified — the normalized package is a separate structured representation.

```
NORMALIZED INPUT PACKAGE contains:
  resume.sections[]         — labelled sections with canonical header names
  resume.bullets[]          — all bullets extracted, tagged EXPLICIT/PROXY/MISSING metric
  resume.dates[]            — all dates in YYYY-MM format
  resume.noise_flags[]      — list of noise terms flagged for candidate review
  resume.consistency_flags[]— gaps, regressions, tenure anomalies
  jd.required[]             — required skill/keyword list
  jd.preferred[]            — preferred skill/keyword list
  jd.inferred[]             — inferred skills from responsibilities section
  jd.level_signals{}        — {years_required, title_seniority, reporting_line}
```

```
jd.kpi_signals[]          — extracted metric vocabulary from JD
jd.quality_flag           — STANDARD | LOW_SIGNAL_JD | AMBIGUOUS_LEVEL
```

> *ASSUMPTION: The normalization rules above are theoretically derived. The proxy metric word list, noise term list, and gap/tenure thresholds are design hypotheses. They become FACT-grounded after 50+ applications per cluster-level provide outcome data on which normalisation decisions improved vs degraded signal accuracy.*

SOURCE: EDLIGO 2025 (parsing failure rates) + Cowan 2001 (cognitive load) + v3.3 Sec 2 (Constraint Engine) + Risk Signal Register (v1.0 Sec 4.2)

3.9 AUTHENTICITY ENGINE [NEW]

> *74% of HR professionals can spot AI-generated resumes. 57% are less likely to hire those applicants (Resumly 2025). The Authenticity Engine is the system's answer to the silent kill pattern: a technically perfect resume that reads like it was assembled by a machine.*

3.9.1 The Optimisation Paradox

There is a tension at the core of any resume optimisation system: the more precisely a system applies structural rules, the more the outputs begin to resemble each other. When every bullet starts with a strong action verb, follows the exact formula, and hits the metric density target — the document is optimally engineered and immediately recognisable as machine-produced.

The Authenticity Engine does not undo the structural work. It introduces controlled, deliberate variation into the language — phrasing diversity, sentence rhythm variation, first-person voice consistency — that preserves signal quality while eliminating the homogenous, over-formulated texture that flags AI generation to experienced recruiters.

SOURCE: Resumly 2025: 74% of HR professionals detect AI-generated resumes. Research Doc Domain 7 (competitor analysis): all current tools optimise for machines at the expense of human authenticity.

3.9.2 Authenticity Engine Functions

Function	Problem It Solves	Rules	What Is NOT Changed
LANGUAGE_VARIATION	Every bullet follows identical structural pattern → reads like a template	1. Cap repetition of any single action verb to maximum 2 occurrences across the full working copy. If Architected appears 4 times → replace 2 with Designed, Engineered, or Built (same level, different word). 2. Vary sentence openings: no more than 3 bullets with the same syntactic structure (Action + Stack + Scale + Metric) in immediate sequence. Break with a Cognitive Signal bullet (problem → action → outcome) to create rhythm variation. 3. Sentence length variation: target a mix of short (10–15 word) and medium (20–30 word) bullets within each role section. Avoid all bullets in a section being the same length.	Signal type (Hard/Cognitive/Structural/Trust) is never changed. Metrics are never removed. The formula is applied — but not applied identically to every bullet in sequence.
VOICE_CONSISTENCY	Resume sounds like different people wrote each section — different levels of formality, different first/third-person usage, inconsistent contraction style	1. Establish voice profile from candidate's master resume: formal/semi-formal/technical. Apply consistently. 2. No mixing of first-person ('I designed') and third-person ('Designed') within the same document — pick one and apply everywhere. Resume OS default: implicit first-person (action verb as opener, no pronoun) unless master resume uses explicit first-person. 3.	Level-appropriate verb hierarchy is not relaxed. Passive openers are still eliminated. Voice consistency is applied ON TOP OF formula requirements.

		Contraction usage: if master resume uses contractions → preserve in summary. If not → exclude throughout. Do not introduce contractions where master resume had none.	
IMPERFECTION_TOLERANCE	Every bullet has a metric → reads like a quota was met, not like real work was described	1. Not every bullet needs a metric IF the preceding bullet in the role already contains a strong Hard Signal. The 'minimum 1 metric per 2 bullets' rule from Engine 5.3 is the floor — do not force metrics into every bullet to exceed this threshold. 2. Allow one bullet per role to be a scope/context bullet: describes the environment, scale of the system, or team structure without a metric. This provides the human reader with context and makes the metricked bullets feel earned rather than formulaic. 3. Do NOT use this tolerance to permit passive language or vague ownership. Scope bullets must still begin with ownership verbs and describe the candidate's domain.	Metric density minimum (1 per 2 bullets) is a floor, not a ceiling. The Authenticity Engine prevents the floor from becoming the ceiling by adding context bullets that give metrics contrast.
NARRATIVE_COHERENCE_CHECK	Bullets within a role have no logical through-line — they read as a list of disconnected achievements rather than a coherent story of what the candidate owned	1. Check that the first bullet of each role establishes the candidate's primary ownership domain. (What did they own? What system or function?) This is the frame. 2. Subsequent bullets should show: depth (how they worked within that domain) + expansion (adjacent areas they touched) + impact (outcomes). 3. If bullets are ordered by metric size (largest metric first for primacy) but create narrative incoherence — log in Strategy Log as NARRATIVE_TENSION. Recommend candidate review to choose between metric-optimised order and narrative-coherent order. System generates both sub-variants.	Primacy/recency rules from Engine 5.3 are not overridden. NARRATIVE_TENSION is a flag, not a block — candidate decides.

3.9.3 Authenticity Score

The Authenticity Engine produces an Authenticity Score (0–100%) reported alongside the other 6 scores in the output package.

```
Authenticity Score =
  Language Variation sub-score (30%):
    verb repetition penalty: -5 pts per verb appearing >2× in working copy
    syntactic pattern repetition: -3 pts per 3+ consecutive same-structure bullets
    sentence length variance: score 0-30 based on std deviation of bullet word counts

  Voice Consistency sub-score (30%):
    person consistency: PASS=30, any violation=0
    formality consistency: -5 pts per section with different formality level

  Imperfection Tolerance sub-score (20%):
    context bullet present: YES=20, NO=0
    no metric-stuffing (>80% bullets with metric): PASS=10, FAIL=0 (add to above)

  Narrative Coherence sub-score (20%):
    first bullet establishes ownership domain: YES=10, NO=0
    logical progression (domain → depth → expansion → impact): PASS=10, FAIL=0
```

```
Thresholds:
  85-100%: Reads as human-authored, engineered resume
  70-84%:  Minor template texture — unlikely to be flagged
  55-69%:  Moderate AI texture — risk of HR detection increasing
  <55%:    Strong AI texture — flag for candidate review before submission
```

> *ASSUMPTION: The Authenticity Score weights are design hypotheses. The '74% HR detection rate' is from Resumly 2025 (single-vendor research — treat as INFERENCE until replicated). The score requires calibration against actual recruiter feedback data via the Failure Diagnosis Layer.*

SOURCE: Resumly 2025 (AI detection rates) + Cowan 2001 (cognitive load) + NNG Bypassing Pattern (repetitive openers) + v3.0 Sec 1.2.5

3.10 TRADE-OFF ENGINE [NEW]

> *The system optimises 7 dimensions simultaneously (ATS, Recruiter Scan, Signal Density, Cognitive Load, Role Match, Simulation, Authenticity). Without explicit conflict resolution rules, these dimensions create contradictions: more keywords improves ATS but degrades readability; more metrics improves Signal Density but can trigger AI detection. This engine makes those trade-offs deterministic.*

3.10.1 The Optimisation Conflict Map

Conflict	Dimension A	Dimension B	How They Conflict	Without Resolution
Keyword density vs readability	ATS Compatibility Score (keywords boost this)	Cognitive Load Score (keyword density degrades this)	Distributing keywords into every bullet increases ATS keyword coverage but creates repetitive, dense text that increases cognitive load and degrades Authenticity Score.	System tries to satisfy both — produces bullets with 3+ keywords per line that read as keyword-stuffed to human reviewers.
Metric density vs authenticity	Signal Density Score (more metrics = higher score)	Authenticity Score (>80% bullets with metrics = metric-stuffing flag)	Engine 5.3 pushes for maximum metric density. Authenticity Engine flags over-metrication as AI texture.	System over-optimises for Signal Density, reducing Authenticity. Resumes pass ATS and fail human review.
Trust Signal prominence vs cognitive load	Simulation Score (Trust Signals in Primary Optical Area = Pedigree Heuristic Score boost)	Cognitive Load Score (too many bold elements = desensitisation)	Bolding all Trust Signals increases Pedigree Heuristic Score but can exceed the 3-bold-per-bullet limit, degrading Cognitive Load Score.	Bold enforcement is applied without regard to Pedigree Heuristic benefit — Trust Signals compete with metrics for bold allocation.
ATS keyword coverage vs authenticity	ATS Compatibility Score	Authenticity Score	Forcing all JD keywords into the resume produces a keyword-dense document that scores well on ATS but reads as unnatural to human reviewers.	High ATS + low Authenticity = likely detected as AI-generated and rejected by human reviewer despite ATS pass.
Senior role framing vs ATS density	Signal Density Score (dense metrics)	Simulation Score — Pedigree Heuristic + Cognitive Load Mode (senior roles require strategic/org-l	For L4+ roles, dense metric bullets signal L3-level execution focus. Strategic framing requires fewer, bigger metrics embedded in system-level language.	System applies same metric density rules to L4 that it applies to L3 — produces mis-levelled output.

		evel language, not dense metrics)		

3.10.2 Trade-Off Resolution Rules — Deterministic

These rules run AFTER all 7 scores are calculated for each variant but BEFORE the default recommendation is finalised. They resolve conflicts by adjusting which score is prioritised in the final output optimisation pass.

```
RULE: ATS_PRIORITY_GATE
  IF ATS Compatibility Score < 70%:
    → Prioritise ATS over Readability, Authenticity, Cognitive Load
    → Keyword Engine may re-run with higher density distribution
    → Authenticity Engine tolerates lower score (floor: 50%) to close ATS gap
    → Flag in Strategy Log: 'ATS Priority Mode active — readability trades accepted'

RULE: HUMAN_PRIORITY_GATE
  IF ATS Compatibility Score >= 80%:
    → Prioritise Recruiter Scan + Cognitive Load + Authenticity over further ATS gain
    → No additional keyword injection beyond current density
    → Authenticity Engine runs at full enforcement
    → Flag in Strategy Log: 'ATS threshold met — human optimisation mode active'

RULE: SENIOR_ROLE_ADJUSTMENT
  IF Role DNA Level = L4 or L5+:
    → Prioritise Trust Signals + Cognitive Signals over Hard Signal density
    → Metric density minimum relaxed: 1 metric per 3 bullets (vs 1 per 2)
    → Bold allocation priority: Trust Signal name first, metric second
    → Summary uses strategic framing, not metric-first framing
    → Flag in Strategy Log: 'Senior role mode — strategic signal weighting applied'

RULE: FRESHER_ADJUSTMENT
  IF Role DNA Level = L1-L2 AND candidate has no work history:
    → Prioritise Trust Signals + Structural Signals over Hard Signal density
    → Proxy metrics accepted at same weight as explicit metrics
    → Assessment score in header block takes priority over company names
    → Authenticity Engine imperfection tolerance raised: floor 2 context bullets per section
    → Flag in Strategy Log: 'Fresher mode — proxy signal weighting active'

RULE: METRIC_STUFFING_BLOCK
  IF Authenticity Score sub-score for Imperfection Tolerance = 0
  AND Signal Density Score > 85%:
    → Authenticity Engine removes lowest-value metric bullets until
       ratio of metricked bullets drops below 80%
    → Removed metric bullets replaced with context/scope bullets
    → Flag in Strategy Log: 'Metric-stuffing correction applied'

RULE: BOLD_ALLOCATION_PRIORITY
  For each bullet where Trust Signal name + metric value BOTH qualify for bold:
    IF cluster Priority 1 = HARD → bold metric value, not Trust Signal name
    IF cluster Priority 1 = TRUST → bold Trust Signal name, not metric value
    IF both are Priority 1-2 → bold metric value (max 3 per bullet enforced)
```

> *These rules are tagged INFERENCE. They are logically derived from the research base and the Signal Architecture priority stacks. They become FACT-grounded once the Failure Diagnosis Layer identifies which trade-off decisions correlate with interview rate differences across the same cluster-level combination.*

3.10.3 Trade-Off Resolution in the Output Package

Every variant's Strategy Log includes a Trade-Off Resolution Log entry: which rules fired, what was adjusted, and what was traded to make the adjustment. This is a transparency output — candidates understand why a high-metric variant may have lower Authenticity, and vice versa.

```
TRADE-OFF LOG FORMAT:
  Rules fired: [list of rule names]
  Adjustments made: [what changed]
  Scores before adjustment: [ATS / Scan / Density / Load / Role / Sim / Auth]
  Scores after adjustment: [same 7 dimensions]
  Net effect: [primary score benefited → secondary score traded]
```

SOURCE: v3.3 Sec 3 (Decision Engine conflict resolution) + Patch C (signal conflict rules) + Resumly 2025 (AI detection) + Cowan 2001 (cognitive load)

3.11 FAILURE DIAGNOSIS LAYER [NEW]

> *Distinct from the Compound Learning System (Section 3.5). The Learning System operates at macro scale — 50+ samples, pattern conclusions, system-wide defaults. The Failure Diagnosis Layer operates at micro scale — per rejected application, per failed engine, per misclassified signal. One is a telescope. The other is a scalpel.*

3.11.1 What Failure Diagnosis Is Not

The Learning System (Section 3.5) identifies what works across populations. The Failure Diagnosis Layer identifies why a specific application failed. A candidate who submits 12 applications, gets 0 responses, and submits all 12 with the same variant has no useful population data for the Learning System — but has 12 diagnosis opportunities.

The layer requires candidate-reported outcome data to function. If the candidate does not report a rejection, the system cannot diagnose. The output package prompt (Section 3.7) explicitly asks for outcome reporting as a required input to the learning system — Failure Diagnosis is the per-application benefit that makes outcome reporting worth the candidate's effort.

3.11.2 Failure Classification Tree

When a candidate reports no response or an early rejection, the system runs the Failure Classification Tree. This is a deterministic diagnostic protocol — not an inference model.

```
FAILURE CLASSIFICATION TREE

INPUT: Application submitted → Outcome: No response OR Rejection at screen

STEP 1: ATS GATE CHECK
  Was ATS Compatibility Score < 70% at submission?
  YES → Diagnosis: ATS_FILTER_FAILURE
         Root cause: keyword coverage gap OR parsing gate issue
         Engine responsible: Engine 5.2 (Keyword) or Engine 5.8 (Structure)
         Action: Re-run keyword engine with higher density; verify parsing gate
  NO → proceed to Step 2

STEP 2: TRIAGE ZONE CHECK
  Was Simulation Score dimension 'Triage Pass Probability' < 70%?
  YES → Diagnosis: TRIAGE_FAILURE
         Root cause: six critical data points not all in top 40%
         Engine responsible: Engine 5.8 (Structure)
         Action: Restructure variant; verify all 6 triage anchors in triage zone
  NO → proceed to Step 3

STEP 3: PEDIGREE SIGNAL CHECK
  Was Simulation Score dimension 'Pedigree Heuristic Score' < 50%?
  YES → Diagnosis: TRUST_SIGNAL_WEAKNESS
```

```
        Root cause: insufficient or mis-placed Trust Signals
        Engine responsible: Engine 5.6 (Signal Amplification) + Engine 0 (classification)
        Action: Review Trust Signal placement; verify company names bold and above fold
        Note: If candidate genuinely has no Trust Signals (no recognisable companies
        or institutions) → flag as STRUCTURAL_TRUST_GAP — cannot be resolved through
        optimisation. Route to candidate development: build verifiable credentials.
  NO → proceed to Step 4

STEP 4: CONFIRMATION BIAS CHECK
  Was Simulation Score dimension 'Confirmation Bias Anchor' = NEGATIVE?
  YES → Diagnosis: NEGATIVE_ANCHOR_FAILURE
        Root cause: first triage-zone element is a risk signal
        Engine responsible: Engine 5.8 (Structure) + Engine 5.9 (Bias Mitigation)
        Action: Move first visible content element to a Hard or Trust Signal
  NO → proceed to Step 5

STEP 5: SIGNAL DENSITY CHECK
  Was Signal Density Score < 60%?
  YES → Diagnosis: SIGNAL_POVERTY
        Root cause: insufficient Hard Signals in experience section
        Engine responsible: Engine 5.3 (Bullet) + Engine 5.6 (Amplification)
        Action: Re-run bullet engine with forced proxy metric where explicit metric absent
        Escalate to candidate: 'Core content gap — develop quantifiable evidence'
  NO → proceed to Step 6

STEP 6: AUTHENTICITY CHECK
  Was Authenticity Score < 60%?
  YES → Diagnosis: AI_TEXTURE_REJECTION
        Root cause: over-optimised language flagged as machine-generated
        Engine responsible: Authenticity Engine (3.9) — was not enforced or
        Trade-Off Engine overrode it (check Trade-Off Log)
        Action: Re-run Authenticity Engine with stricter enforcement;
        review Trade-Off Log for which rules suppressed Authenticity
  NO → proceed to Step 7

STEP 7: LEVEL MISMATCH CHECK
  Did candidate target L4+ role with Role Match Score < 65%?
  YES → Diagnosis: LEVEL_MISMATCH_FAILURE
        Root cause: functional experience does not meet JD requirements at target level
        Engine responsible: Engine 5.4 (Role-Shift) — may have language-shifted
        without underlying evidence
        Action: Escalate to candidate: 'Evidence gap at target level —
        language optimisation cannot close this gap'
  NO → Diagnosis: UNCLASSIFIED_FAILURE
        All diagnosed dimensions pass. Failure may be: market conditions,
        role filled internally, budget freeze, or pure recruiter variance.
        Action: Log as UNCLASSIFIED. Continue with next application.
        Do not re-optimise without new signal.
```

3.11.3 Failure Diagnosis Output

The Failure Diagnosis output is appended to the original Strategy Log for the failed application. It is not a new document — it is a diagnostic addendum.

```
FAILURE DIAGNOSIS ADDENDUM:
  Application ID: [unique run ID]
  Outcome reported: [No response | Screen rejection | Interview rejection]
  Diagnosis: [FAILURE_TYPE from classification tree]
  Root cause: [specific engine + rule that produced the failure]
  Actionable fix: [re-run instruction OR candidate development escalation]
```

```
  Resolvable by optimisation: [YES | NO | PARTIAL]
  Recommended next action: [specific instruction]
```

'Resolvable by optimisation' is the critical field. If the answer is NO (e.g., STRUCTURAL_TRUST_GAP or LEVEL_MISMATCH_FAILURE), the system does not recommend re-running Resume OS. It routes the candidate to the correct intervention: build verifiable credentials, develop quantifiable project outcomes, or adjust target role level. Resume OS cannot fix a content problem.

SOURCE: v1.0 Sec 9 (learning system) + Simulation Layer Section 3.4 + Risk Signal Register v1.0 Sec 4.2 + Interviewing.io 2024 (recruiter variance)

3.12 VARIANT DISTANCE CONTROL [NEW]

> *Generating 10 variants that are cosmetically different — same content, slightly different section order — is not a variant strategy. It is testing the same hypothesis 10 times. Variant Distance Control enforces that each variant differs in at least one of three substantive dimensions: structure, signal emphasis, or pattern strategy.*

3.12.1 The Variant Redundancy Problem

Without distance enforcement, the Output Engine (Layer 6) can generate variants that share the same structure (Experience first) with minor reorderings that do not change what the recruiter sees in the triage scan. The learning system then receives 10 data points for what is effectively 1 hypothesis — corrupting pattern analysis.

3.12.2 The Three Distance Dimensions

Dimension	What It Measures	Minimum Distance Requirement	How It Is Measured
Structure Distance	Difference in section ordering between two variants	At least one section-level difference: Education position, Skills position, or Summary inclusion/exclusion must differ between any two variants	Compare section order arrays. If identical → variants are structurally the same. Merge or regenerate one.
Signal Emphasis Distance	Difference in which signal type is foregrounded in the triage zone (top 40%)	At least two variants must differ in whether their triage zone leads with a Hard Signal or a Trust Signal as the dominant element	Compare the signal type of the first non-contact element in the triage zone. If all variants lead with the same signal type → at least one variant must be restructured to lead with the next-priority signal type.
Pattern Strategy Distance	Difference in the scanning pattern the variant is designed to activate (V1–V10)	No two variants in the same output package may use the same base variant template (V1–V10)	Compare variant template assignment. If duplicate template assigned → the lower-scoring duplicate is removed and a different template assigned to maintain minimum 7.

3.12.3 Minimum Distance Threshold — The Delta Rule

For any pair of variants in the output package, the Variant Distance Score must be ≥ 1. If any pair scores 0 (no dimension differs) → the pair is redundant and one must be regenerated.

```
VARIANT DISTANCE SCORE for pair (Vi, Vj):
  +1 if Structure Distance > 0 (section order differs)
  +1 if Signal Emphasis Distance > 0 (triage zone lead signal type differs)
  +1 if Pattern Strategy Distance > 0 (different V1-V10 template)

  Minimum acceptable: Variant Distance Score >= 1
  Ideal: Variant Distance Score = 3 (differs on all three dimensions)
```

```
ENFORCEMENT:
  Before delivery: compare all variant pairs
  If any pair scores 0 → flag as REDUNDANT_VARIANTS
  Action: regenerate the lower-scoring variant of the pair
  with a different template that creates distance on at least 1 dimension
```

3.12.4 Variant Coverage Map — What the 7 Minimum Variants Should Cover

To maximise hypothesis diversity across the output package, the minimum 7 variants should collectively cover the following combinations. This is a recommendation, not a hard constraint — Role DNA and ATS detection may make some combinations invalid for specific runs.

Variant Slot	Priority Coverage	Template Target	Signal Lead
Slot 1	Primary ATS-optimised	V1 or V2 (matched to cluster)	Hard Signal leads triage zone
Slot 2	Primary human-optimised	V2, V4, or V6 (matched to level)	Trust Signal or Cognitive Signal leads triage zone
Slot 3	ATS platform-specific alternate	V3 or V9 (MNC/enterprise format)	Structural Signal — compliance-first framing
Slot 4	Level-specific emphasis	V4 (L4+) or V1 (L1-L2)	Matches Level axis signal priority
Slot 5	Environment variant	Same as Slot 1 but with opposite Environment adaptation applied	Same signal lead as Slot 1, different tone
Slot 6	Transition or breadth	V8 or V10 (versatility or pivot)	Hard Signal with environment amplification
Slot 7	High authenticity	Any template with Authenticity Engine at max enforcement	Mixed signal lead — narrative coherence prioritised

`SOURCE: v1.0 Sec 7 (variant model) + v2.0 Sec 8.2 (10 base variants) + v3.3 Sec 6 + Compound Learning System Section 3.5`

3.13 FALLBACK MODES & DEGRADED OPERATION [NEW]

> *The system is specified for ideal inputs: structured master resume, complete JD, detected ATS. Real usage breaks these assumptions. Fallback Modes define how Resume OS behaves when inputs are incomplete, ambiguous, or absent — and what the system tells the candidate about the quality degradation.*

3.13.1 Fallback Mode Table

Trigger Condition	Mode Name	System Behaviour	Capability Degradation	Candidate Notification
No JD provided — candidate submits resume only, no target role	BLIND_MODE	1. Infer Role DNA from resume: most recent title → Function axis. Years of experience → Level axis. Company size from most recent role if detectable → Environment axis. 2. Generate generic 3 variants (V1 Signal Stack, V2 Outcome Ledger, V4 Leadership Thesis) without role-specific	ATS Compatibility Score: NOT CALCULATED (no JD). Role Match Score: NOT CALCULATED. Gap Notification: NOT AVAILABLE. Default	'Resume optimised for your experience profile and inferred role cluster. For role-specific optimisation with ATS compatibility scoring, provide a target job description or URL.'

		optimisation. 3. Apply generic ATS-safe rules (single column, DOCX, standard headers). 4. Keyword Engine skips — no JD to extract from. 5. Gap Scorecard skips — no JD to gap against.	Recommendation: 'Generic variant — no role target provided'.	
ATS URL not detected or JD text only (no URL)	GENERIC_ATS_MODE	1. Apply universal ATS-safe rules: single column, DOCX format, standard headers, exact tool names, no tables/icons. 2. ATS Compatibility Score calculated against generic keyword coverage model (not platform-specific). 3. Flag: 'Unknown ATS — platform-specific rules not applied'. 4. All other layers run normally.	ATS Compatibility Score accuracy reduced: platform-specific keyword weighting not applied. File format recommendation: DOCX (safe default). Semantic variant expansion: disabled (no ATS type to select for).	'ATS platform not detected. Generic formatting applied. For platform-specific optimisation, provide the job application URL.'
Master resume has no quantifiable metrics — SIGNAL_POVERTY flag from Input Normalization	PROXY_ENFORCED_MODE	1. Bullet Engine forced to apply proxy metric extraction to every bullet. 2. Scope indicators (team size, geographic scope, volume words) treated as proxy metrics at full weight for Signal Density calculation. 3. Strategy Log flags every proxy metric: 'PROXY — explicit metric unavailable'. 4. Candidate notified that proxy metrics are a temporary measure and coached on developing explicit metrics for 3 high-priority bullets.	Signal Density Score calculated using proxy metrics — score is valid but carries lower human review weight than explicit metrics. Simulation Score Pedigree Heuristic not affected. Hard Signal sub-score capped at 70% when >50% of metrics are proxies.	'Your resume currently has limited quantifiable outcomes. Proxy signals (team size, system scale) have been applied. To strengthen interview conversion, develop explicit metrics for your top 3 achievements. The Strategy Log identifies which bullets would benefit most.'
JD has <5 required skills (LOW_SIGNAL_JD flag from Input Normalization)	LOW_SIGNAL_JD_MODE	1. Keyword Engine expands to Inferred skills list (from JD responsibilities section) for keyword extraction. 2. Role DNA Level and Environment axes inferred from title and company signals in the JD. 3. Gap Scorecard keyword coverage target reduced to 50% threshold (vs 70% standard) — JD simply does not have enough keywords to hit 70%. 4. ATS Advisory includes: 'This JD is low-signal. Keyword optimisation is limited by JD content.'	Keyword Coverage%: threshold lowered to 50% before BLOCK triggers. ATS Compatibility Score accuracy reduced. Role Match Score: INFERRED (partial).	'This job description has limited keyword signals. Optimisation has been applied to available signals. If you have access to additional role information (internal job spec, similar role examples), adding them would improve accuracy.'
Candidate submits a 10+ page master resume or a single-paragraph bare-bones resume	CONTENT_VOLUME_CORRECTION	1. Long resume (10+ pages): Recency Weighting Engine applies maximum compression immediately. Roles beyond 15 years compressed to title + company + dates. Candidate notified before output is generated that significant compression will occur. 2.	Long resume: output quality is normal — compression is a feature. Bare-bones: output quality is significantly degraded. System	Long: 'Significant compression applied to pre-[year] experience. Review Strategy Log for what was condensed.' Bare-bones: 'Your master resume has insufficient content for role-specific optimisation. The Gap

		Bare-bones resume (<200 words): system flags CONTENT_POVERTY. Generates a Gap Amplification Report: identifies the 5 highest-impact additions the candidate could make to their master resume before re-running Resume OS.	recommends against submitting until Gap Amplification is addressed.	Amplification Report identifies the 5 additions that would most improve your signal. Resume OS can be re-run after additions.'
Application timing: posting is > 10 days old	LATE_APPLICATION_WARNING	1. ATS Advisory appended with timing signal: posting age + interview probability reduction. 2. No change to optimisation logic — document quality optimisation runs normally. 3. Strategy Log note: 'Late application: TalentWorks research indicates interview probability drops to ~1.5% after 10 days. Consider applying immediately and simultaneously developing relationships at the company via other channels.'	No capability degradation. Warning only.	'This posting is [N] days old. Research indicates interview probability significantly decreases after 10 days. Submit promptly. The resume is optimised regardless of timing.'

`SOURCE: v1.0 Sec 10 (execution protocol) + v3.3 Sec 2 (Constraint Engine) + TalentWorks application timing data + Input Normalization Engine Section 3.8`

GLOSSARY — CANONICAL DEFINITIONS

All terms used in this document. These are the canonical definitions. Any external documentation should reference these — do not re-define independently.

Role DNA: Structured classification: Function + Level + Environment + KPIs. Every run begins with this. Format: [Function] + [Level] + [Environment] + [KPIs].

Hard Signal: 40% of Signal Architecture. Quantified, verifiable outcome. Activates Spotted Pattern. Cannot be simulated. Examples: quota%, ARR, model accuracy vs baseline, latency delta.

Cognitive Signal: 25% of Signal Architecture. Problem→Diagnosis→Action→Outcome bullet structure. NOT claimed — demonstrated. Does not directly improve ATS score.

Structural Signal: 20% of Signal Architecture. Conveyed by resume architecture. Determines Parsing Gate pass/fail. Cannot be engineered through content — only through formatting discipline.

Trust Signal: 15% of Signal Architecture. Third-party credibility. Activates pedigree heuristic. Does NOT improve ATS score — for human cognitive processing only.

Parsing Gate: Binary pass/fail structural compliance check. Failure invalidates all ATS score components regardless of content quality.

Triage Zone: Top ~40% of page one. Contains all six critical fixation data points. Optimisation target for 7–31 second triage window.

Layer-Cake Pattern: NNG-documented optimal scanning pattern. Target state for all Resume OS outputs. Activated when section headers are visually prominent information carriers.

Spotted Pattern: NNG-documented pre-attentive fixation on bold text and numerals. Primary mechanism by which metrics intercept the triage scan.

F-Pattern: Failure-state scanning pattern from poor document structure. Resume OS engineers documents to ESCAPE F-pattern and activate Layer-Cake.

Bypassing Pattern: NNG-documented: readers skip first 2–3 words of each line when repetitive or predictable. 'Responsible for...' triggers Bypassing Pattern.

Proxy Signal Architecture: Framework for generating career signal for freshers with no work history. 6 proxy types substitute for traditional work-experience signals.

Pedigree Heuristic: Documented recruiter behaviour (Interviewing.io 2024): institutional affiliation is the strongest actual predictor of interview selection — overriding stated criteria. FAANG = 35% lift.

Signal Density Score: Composite of 4 sub-scores (Hard 40% + Cognitive 25% + Structural 20% + Trust 15%). Measures verifiable proof per unit of resume content.

Simulation Score: New in vFinal Unified. Models recruiter decision layer: Triage Pass Probability (35%) + Pedigree Heuristic (30%) + Confirmation Bias Anchor (20%) + Cognitive Load Mode (15%) minus risk signal penalties. Tagged INFERENCE — requires outcome data to validate weights.

Confirmation Bias Anchor: First substantive content element visible to recruiter creates interpretive frame for entire document (Asch 1946 + Rabin & Schrag 1999). POSITIVE anchor = Hard or Trust Signal first in Triage Zone.

Cognitive Load Decision Mode: Simulation dimension: predicts whether resume design keeps recruiter in analytical (System 2) vs heuristic (System 1) mode. System 1 = 55% accuracy ceiling.

Decision Engine: Layer 3, Section 3.2+. Translates Role DNA into complete structural prescription. Output is deterministic given inputs. No subjective judgment after Role DNA is classified.

Compound Intelligence: Learning principle: 50-sample minimum per cluster-level before pattern conclusions modify system defaults. Before threshold: ASSUMPTION. After threshold: FACT.

Bias Mitigation Engine: Sub-Engine 5.9. Gender-neutral language, gap-neutral date format, assessment-forward placement for freshers. Each intervention requires candidate consent or opt-in.

India ATS Stack: Naukri RMS, Darwinbox, Keka, Freshteam, GreytHR. All require exact-match keyword optimisation. Naukri RMS dominant for campus hiring.

Master Resume: Candidate's complete, factual, unedited work history. Source of truth. Never overwritten. All outputs generated from working copies.

Three-Window Model: vFinal temporal model: 100ms (pre-attentive), 7–15s (triage), 25–45s (commitment). Replaces 6-second model from v1/v2.

Strategy Log: Required output of every valid run. Documents every change (before/after + rationale), all keyword gaps, and variant recommendation with rationale.

Input Normalization Engine: Pre-processing gate that runs before Engine 0. Converts raw inputs into Normalized Input Package: labelled sections, metric-tagged bullets (EXPLICIT/PROXY/MISSING), structured JD signals, noise flags, consistency flags.

Authenticity Engine: Post-optimisation layer introducing controlled language variation to prevent AI-texture detection. Four functions: Language Variation, Voice Consistency, Imperfection Tolerance, Narrative Coherence. Produces Authenticity Score.

Authenticity Score: 7th score in output package. Composite: Language Variation 30% + Voice Consistency 30% + Imperfection Tolerance 20% + Narrative Coherence 20%. Below 55% = strong AI texture. Tagged INFERENCE pending calibration.

Trade-Off Engine: Conflict resolution layer running after all 7 scores calculated. Rules: ATS_PRIORITY_GATE, HUMAN_PRIORITY_GATE, SENIOR_ROLE_ADJUSTMENT, FRESHER_ADJUSTMENT, METRIC_STUFFING_BLOCK, BOLD_ALLOCATION_PRIORITY. Produces Trade-Off Log.

Failure Diagnosis Layer: Per-application diagnostic protocol. 7-step classification tree: ATS_FILTER_FAILURE → TRIAGE_FAILURE → TRUST_SIGNAL_WEAKNESS → NEGATIVE_ANCHOR_FAILURE → SIGNAL_POVERTY → AI_TEXTURE_REJECTION → LEVEL_MISMATCH_FAILURE → UNCLASSIFIED. Outputs Failure Diagnosis Addendum.

Variant Distance Control: Enforces variant diversity. Three dimensions: Structure (section order), Signal Emphasis (triage zone lead), Pattern Strategy (V1-V10 template). Min Variant Distance Score >= 1 for any pair. Redundant pairs regenerated.

Fallback Modes: Six degraded-operation modes: BLIND_MODE (no JD), GENERIC_ATS_MODE (no URL), PROXY_ENFORCED_MODE (no metrics), LOW_SIGNAL_JD_MODE (<5 skills), CONTENT_VOLUME_CORRECTION (extreme length), LATE_APPLICATION_WARNING (>10 days posting).